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A Simulation-Based Model Design for High-Performance 5G

Senior Project Report

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List of Contents

Contents

Abstract V

Arabic Abstract IV

Acknowledgments IV

List of Contents I

List of Abbreviations VI

Chapter 1 5G Networks 1

1.1 Introduction to 5G 2

1.2 Differences between the previous generations and 5G..... 2

1.2.1 The difference between 4G and 5G 3

1.3 5G use cases 4

1.4 5G Networks Architecture 6

1.5 Technologies used in 5G Networks 8

1.5.1 Millimeter Waves 8

1.5.2 Small Cells 8

1.5.3 Massive MIMO..... 9

1.5.4 Beamforming 10

1.5.5 Full Duplex 11

1.5.6 Non-Orthogonal Multiple Access (NOMA) 12

1.5.7 Network Slicing 14

1.5.8 Infrastructure Based on NFV and SDN.....	15
1.5.9 Multi-Tenancy.....	16
Chapter 2 5G Virtual Network	17
2.1 The concept of Network Slicing.....	18
2.2 Network Slicing Principles	21
2.2.1 Slice Isolation.....	21
2.2.2 Elasticity.....	21
2.2.3 End-to-End Customization.....	22
2.3 Network Slicing Enablers	22
2.3.1 Software Defined Networking.....	22
2.3.2 Network Function Virtualization	24
2.3.3 Cloud Computing.....	25
2.3.4 Edge Computing	27
Chapter 3 Resource Allocation	28
3.1 General introduction	29
3.2 Resource allocation in cloud radio access 5G network	29
3.2.1 Introduction.....	29
3.2.2 The resource allocation element in a CRAN.....	30
3.2.3 Emerging use-cases with the CRAN architecture	31
3.3 Resource allocation in D2D.....	34
3.4 Resource allocation in virtual networks	35
3.4.1 Resource allocation in network slicing	36
Chapter 4 Optimization	39

4.1 Optimization Problem Types Overview	40
4.2 Optimizing Problem	43
Chapter 5 Network Simulation Using OMNeT++	49
5.1 Introduction to OMNeT++	50
5.2 Setting up the project in OMNeT++	52
5.3 Adding omnetpp.ini file	52
5.4 Adding the Network Description File (NED)	54
5.5 Adding the C++ files	Error! Bookmark not defined.
5.7 Our proposal scenarios for the network simulation	55

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Abstract

With the expansion and proliferation of 5G networks and with their support for many services besides calls and Internet services as their support for Internet of Things services, there has become a significant burden on the network's resources and this was one of the important challenges that 5G networks faces. Therefore, it became necessary to improve the proper use and distribution of these resources to ensure that they are not wasted, consumed or employed incorrectly. Hence, we started with the idea of our project to build a model to optimize resource allocation in virtual 5G networks. Through our project proposal, the statue of allocating a network resource will be discussed by converting these resources into mathematical equations and finding optimal solutions to these equations and then simulating these solutions to see how they affect network resources and the quality of communication between the user and the network.

List of Abbreviations

1G	First Generation
2G	Second Generation
3G	Third Generation
4G	Fourth Generation
LTE	Long Term Evolution
5G	Fifth Generation
IoT	Internet of things
TDMA	Time Division Multiple Access
CDMA	Code Division Multiple Access
MIMO	Multiple Input Multiple Output
NVF	Network Function Virtualization
NOMA	Non-Orthogonal Multiple Access
OMA	Orthogonal Multiple Access
KPI	Key Performance Indicator
SDN	Software Defined Network
NFV	Network Function Virtualized
RAN	Radio Access Network
VNF	Virtual Network Functions
mMTC	Massive Machine Type Communications
MANO	Management And Orchestration
CMBB	Mobile Broad Band
CDN	Code Defined Network
NAT	Network Address Translation
VPN	Virtual Private Network
IAAS	Infrastructure As a Service
PAAS	Platform As Service
SAAS	Software As Service
ETSI	European Telecommunications Standards Institute
CRAN	Cloud Radio Access Network

RRH	Remote Radio Heads
BBU	Base Band Unit
QoS	Quality Of Service
QoE	Quality Of Experience
mmWave	Millimeter wave
APs	BSS/access points
D2D	Device-To-Device
FD	Full Duplex
SI	Self-Interference
MNOS	Mobile Network Operator
OMA	Orthogonal Multiple Access
OFDMA	Orthogonal Frequency Division Multiple Access
RBs	Resource Blocks
NS	Network Slicing
E2E	End To End
VNUs	Virtual Network Units
DCs	Data Centers
EE	Energy Efficiency
VM	Virtual Machine
NED	Network Description File
MU	Mobile User

Chapter 1

5G Networks

1.1 Introduction to 5G

5G is the 5th generation of mobile networks, a significant evolution of today's 4G LTE networks. 5G has been designed to meet the very large growth in data and connectivity of today's modern society, the internet of things (IoT) with billions of connected devices, and tomorrow's innovations. 5G will initially operate in conjunction with existing 4G networks before evolving to fully standalone networks in subsequent releases and coverage expansions. In addition to delivering faster connections and greater capacity, a very important advantage of 5G is the fast response time referred to as latency. Latency is the time taken for devices to respond to each other over the wireless network. 3G networks had a typical response time of 100 milliseconds, 4G is around 30 milliseconds and 5G will be as low as 1 millisecond. This is virtually instantaneous opening up a new world of connected applications [1].

1.2 Differences between the previous generations and 5G

The previous generations of mobile networks are 1G, 2G, 3G, and 4G.

- First generation-1G: 1980s, 1G delivered analog voice.
- Second generation - 2G: Early 1990s, 2G introduced digital voice (e.g., - Time Division Multiple Access (TDMA)).
- Third generation - 3G: Early 2000s, 3G brought mobile data (e.g., - Code Division Multiple Access (CDMA)).
- Fourth-generation - 4G LTE: 2010s, 4G LTE ushered in the era of mobile broadband.

1G, 2G, 3G, and 4G all led to 5G, which is designed to provide more connectivity than was ever available before. 5G is a unified, more capable air interface. It has been designed with an extended capacity to enable next-generation user experiences, empower new deployment models, and deliver new services. With high speeds, superior reliability, and negligible latency,

5G will expand the mobile ecosystem into new realms. 5G will impact every industry, making safer transportation, remote healthcare, precision agriculture, digitized logistics, and more a reality [3].

1.2.1 The difference between 4G and 5G

here are several reasons that 5G will be better than 4G:

- 5G is significantly faster than 4G Up to 10Gbps data rate - > 10 to 100x speed improvement over 4G and 4.5G networks.
- 5G has more capacity than 4G: 5G is designed to support a 100x increase in traffic capacity and network efficiency.
- 5G has significantly lower latency than 4G estimated at 1-millisecond latency.
- 5G is a unified platform that is more capable than 4G: While 4G LTE focused on delivering much faster mobile broadband services than 3G, 5G is designed to be a unified, more capable platform that not only elevates mobile broadband experiences but also supports new services such as mission-critical communications and the massive IoT. 5G can also natively support all spectrum types (licensed, shared, unlicensed) and bands (low, mid, high), a wide range of deployment models (from traditional macro-cells to hotspots), and new ways to interconnect (such as device-to-device and multi-hop mesh).
- 5G uses spectrum better than 4G: 5G is also designed to get the most out of every bit of spectrum across a wide array of available spectrum regulatory paradigms and band from low bands below 1 GHz, to mid bands from 1 GHz to 6 GHz, to high bands known as millimeter wave (mmWave).
- 90% reduction in network energy usage.

1.3 5G use cases

5G will enable instantaneous connectivity to billions of devices, the Internet of Things (IoT), and a truly connected world. 5G will provide the speed, low latency, and connectivity to enable a new generation of applications, services, and business opportunities that have not been seen before. There are three major categories of use case for 5G and also as shown in figure (1.1):

- 1) Massive machine to machine communications: also called the Internet of Things (IoT) that involves connecting billions of devices without human intervention at a scale not seen before. This has the potential to revolutionize modern industrial processes and applications including agriculture, manufacturing, and business communications.
- 2) Ultra-reliable low latency communications: mission-critical including real-time control of devices, industrial robotics, vehicle to vehicle communications and safety systems, autonomous driving, and safer transport networks. Low latency communications also open up a new world where remote medical care, procedures, and treatment are all possible
- 3) Enhanced mobile broadband: providing significantly faster data speeds and greater capacity keeping the world connected. New applications will include fixed wireless internet access for homes, outdoor broadcast applications without the need for broadcast vans, and greater connectivity for people on the move.

For communities, 5G will enable the connection of billions of devices for our smart cities, smart schools, and smart homes, smart and safer vehicles, enhance health care and education and provide a safer and more efficient place to live. For businesses and industry, 5G and IoT will provide a wealth of data allowing them to gain insights into their operations like never before. Businesses will operate and make key decisions driven by data, innovate in agriculture, smart farms, and manufacturing, paving the way for cost savings, better customer experience, and long-term growth. New and Emerging technologies such as virtual and augmented reality will be accessible by everyone. Virtual reality provides connected experiences that were not possible before. With 5G and VR, you will be able to travel to your favorite city, watch a live football match with the feeling of being on the ground, or even be able to inspect real estate and walk through a new home all from the comfort of your couch. 5G will keep us connected in tomorrow's smart cities, smart homes, and smart schools, and enable opportunities that we haven't even thought of yet [1].

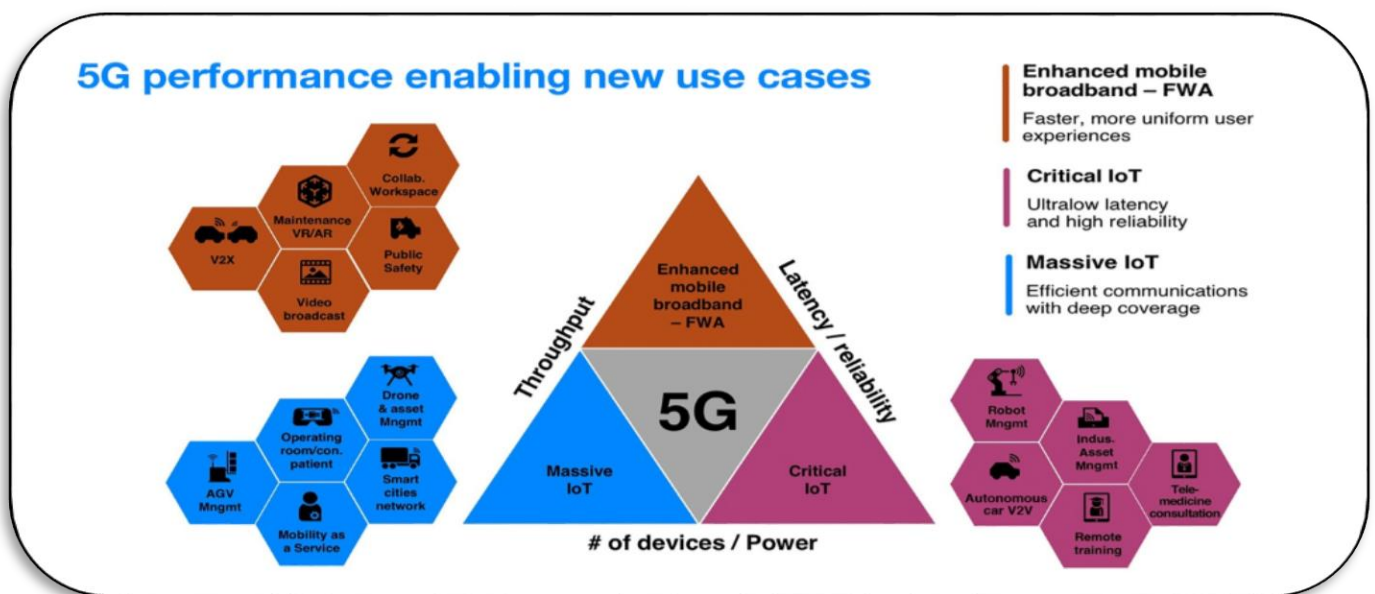


Figure 1.1 Major categories of 5G [2]

1.4 5G Networks Architecture

Most operators will initially integrate 5G networks with existing 4G networks to provide a continuous connection. 5G network architecture illustrating 5G and 4G working together, with central and local servers providing faster content to users and low latency applications. A mobile network has two main components, the ‘Radio Access Network’ and the ‘Core Network’.

The Radio Access Network: consists of various types of facilities including small cells, towers, masts, and dedicated in-building and home systems that connect mobile users and wireless devices to the main core network. Small cells will be a major feature of 5G networks particularly at the new millimeter wave (mmWave) frequencies where the connection range is very short. To provide a continuous connection, small cells will be distributed in clusters depending on where users require a connection which will complement the macro network that provides wide-area coverage. 5G Macro Cells will use MIMO (multiple inputs, multiple outputs) antennas that have multiple elements or connections to send and receive more data simultaneously. The benefit to users is that more people can simultaneously connect to the network and maintain high throughput. Where MIMO antennas use very large numbers of antenna elements they are often referred to as ‘massive MIMO’, however, the physical size is similar to existing 3G and 4G base station antennas.

The Core Network: is the mobile exchange and data network that manages all of the mobile voice, data, and internet connections. For 5G, the ‘core network’ is being redesigned to better integrate with the internet and cloud-based services and also includes distributed servers across the network improving response times (reducing latency). Many of the advanced features of 5G including network function virtualization and network slicing for different applications and services, will be managed in the core. Figure (1.2) illustrates examples of local cloud servers

providing faster content to users (movie streaming) and central server with 4G macro cells and 5G small cells.

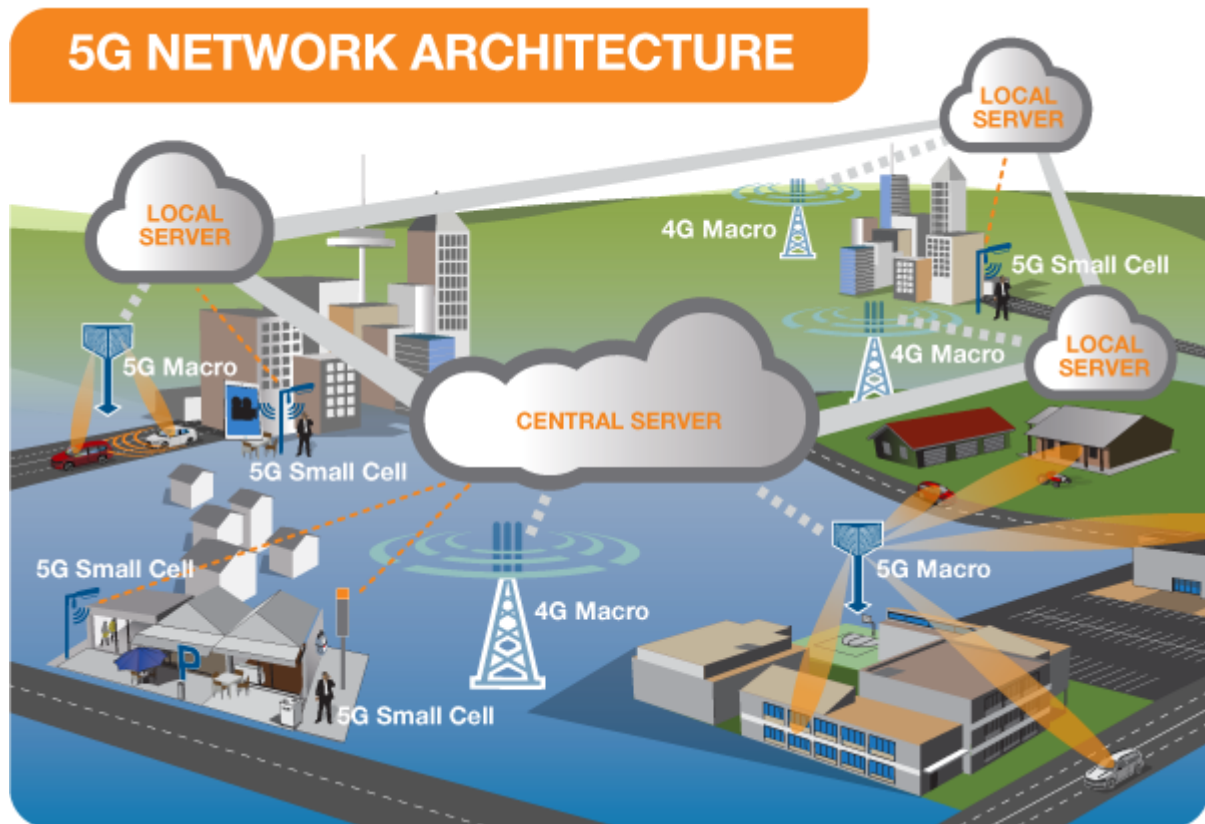


Figure 0.2 5G Architecture [1]

Network Slicing: enables a smart way to segment the network for a particular industry, business, or application. For example, emergency services could operate on a network slice independently from other users.

Network Function Virtualization (NVF): is the ability to instantiate network functions in real-time at any desired location within the operator's cloud platform. Network functions that used to run on dedicated hardware for example a firewall and encryption at business premises can now operate on software on a virtual machine. NVF is crucial to enable the speed efficiency and agility to support new business applications and is an important technology for a 5G ready core. [1]

1.5 Technologies used in 5G Networks

1.5.1 Millimeter Waves

Today's wireless networks have run into a problem: More people and devices are consuming more data than ever before, but it remains crammed on the same bands of the radio-frequency spectrum that mobile providers have always used. That means less bandwidth for everyone, causing slower service and more dropped connections. One way to get around that problem is to simply transmit signals on a whole new swath of the spectrum, one that's never been used for mobile service before. That's why providers are experimenting with broadcasting on millimeter waves, which use higher frequencies than the radio waves that have long been used for mobile phones. Millimeter waves are broadcast at frequencies between 30 and 300 gigahertz, compared to the bands below 6 GHz that were used for mobile devices in the past. They are called millimeter waves because they vary in length from 1 to 10 mm, compared to the radio waves that serve today's smartphones, which measure tens of centimeters in length. Until now, only operators of satellites and radar systems used millimeter waves for real-world applications. Now, some cellular providers have begun to use them to send data between stationary points, such as two base stations. But using millimeter waves to connect mobile users with a nearby base station is an entirely new approach. There is one major drawback to millimeter waves, though they can't easily travel through buildings or obstacles and they can be absorbed by foliage and rain. That's why 5G networks will likely augment traditional cellular towers with another new technology, called small cells [4].

1.5.2 Small Cells

Small cells are portable miniature base stations that require minimal power to operate and can be placed every 250 meters or so throughout cities. To prevent signals from being dropped, carriers could install thousands of these stations in a city to form a dense network that acts

like a relay team, receiving signals from other base stations and sending data to users at any location. While traditional cell networks have also come to rely on an increasing number of base stations, achieving 5G performance will require an even greater infrastructure. Luckily, antennas on small cells can be much smaller than traditional antennas if they are transmitting tiny millimeter waves. This size difference makes it even easier to stick cells on light poles and atop buildings. This radically different network structure should provide more targeted and efficient use of spectrum. Having more stations means the frequencies that one station uses to connect with devices in one area can be reused by another station in a different area to serve another customer. There is a problem, though the sheer number of small cells required to build a 5G network may make it hard to set up in rural areas. In addition to broadcasting over millimeter waves, 5G base stations will also have many more antennas than the base stations of today's cellular networks to take advantage of another new technology: massive Multiple Inputs Multiple Outputs (MIMO).

1.5.3 Massive MIMO

Today's 4G base stations have a dozen ports for antennas that handle all cellular traffic: eight for transmitters and four for receivers. But 5G base stations can support about a hundred ports, which means many more antennas can fit on a single array. That capability means a base station could send and receive signals from many more users at once, increasing the capacity of mobile networks by a factor of 22 or greater. This technology is called massive MIMO describes wireless systems that use two or more transmitters and receivers to send and receive more data at once. Massive MIMO takes this concept to a new level by featuring dozens of antennas on a single array. MIMO is already found on some 4G base stations. But so far, massive MIMO has only been tested in labs and a few field trials. In early tests, it has set new records for spectrum efficiency, which is a measure of how many bits of data can be transmitted to a certain number of users per second. Massive MIMO looks very promising for the future of 5G. However,

installing so many more antennas to handle cellular traffic also causes more interference if those signals cross. That's why 5G stations must incorporate beamforming [4].

1.5.4 Beamforming

Beamforming is a traffic-signaling system for cellular base stations that identifies the most efficient data-delivery route to a particular user, and it reduces interference for nearby users in the process. Depending on the situation and the technology, there are several ways for 5G networks to implement it. Beamforming can help massive MIMO arrays make more efficient use of the spectrum around them as shown in figure (1.3).

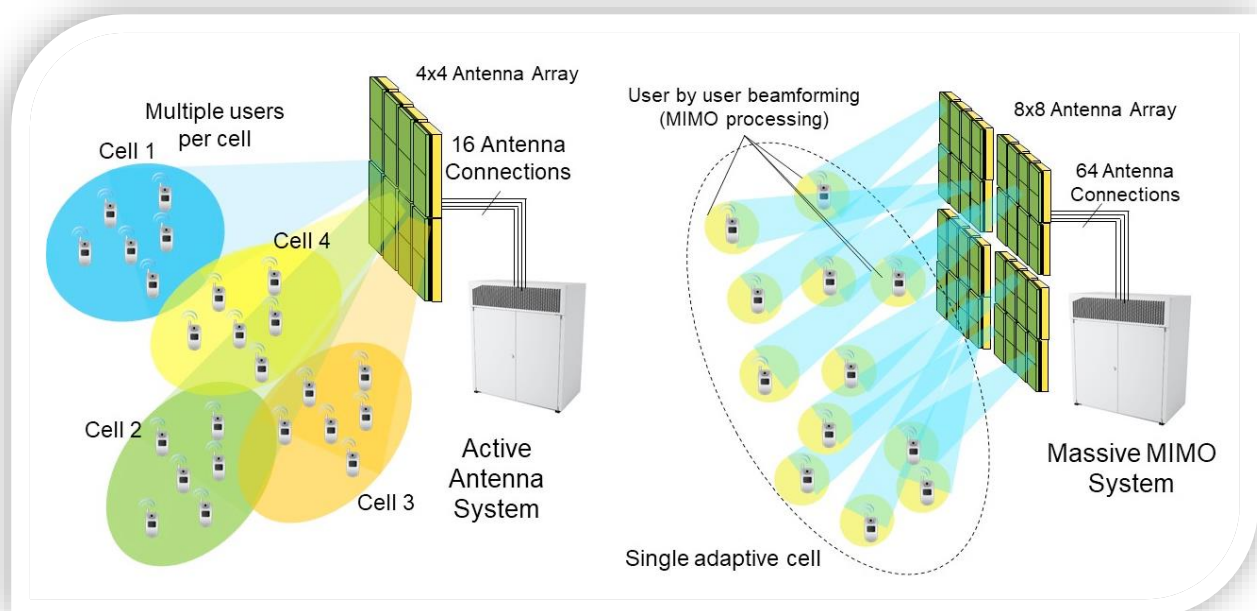


Figure 1.3 Massive MIMO & Beamforming

The primary challenge for massive MIMO is to reduce interference while transmitting more information from many more antennas at once. At massive MIMO base stations, signal-processing algorithms plot the best transmission route through the air to each user. Then they can send individual data packets in many different directions, bouncing them off buildings and other objects in a precisely coordinated pattern. By choreographing the packets' movements and arrival time, beamforming allows many users and antennas on a massive MIMO array to exchange much more information at once. For millimeter waves, beamforming is primarily

used to address a different set of problems: Cellular signals are easily blocked by objects and tend to weaken over long distances. In this case, beamforming can help by focusing a signal in a concentrated beam that points only in the direction of a user, rather than broadcasting in many directions at once. This approach can strengthen the signal's chances of arriving intact and reduce interference for everyone else. Besides boosting data rates by broadcasting over millimeter waves and beefing up spectrum efficiency with massive MIMO, wireless engineers are also trying to achieve the high throughput and low latency required for 5G through a technology called full-duplex, which modifies the way antennas deliver and receive data [4].

1.5.5 Full Duplex

Today's base stations and cellphones rely on transceivers that must take turns if transmitting and receiving information over the same frequency, or operate on different frequencies if a user wishes to transmit and receive information at the same time. With 5G, a transceiver will be able to transmit and receive data at the same time, on the same frequency. This technology is known as full-duplex, and it could double the capacity of wireless networks at their most fundamental physical layer: Picture two people talking at the same time but still able to understand one another—which means their conversation could take half as long and their next discussion could start sooner. Some militaries already use full-duplex technology that relies on bulky equipment. To achieve full-duplex in personal devices, researchers must design a circuit that can route incoming and outgoing signals so they don't collide while an antenna is transmitting and receiving data at the same time. This is especially hard because of the tendency of radio waves to travel both forward and backward on the same frequency a principle known as reciprocity. But recently, experts have assembled silicon transistors that act like high-speed switches to halt the backward roll of these waves, enabling them to transmit and receive signals on the same frequency at once. One drawback to full-duplex is that it also creates more signal interference, through a pesky echo. When a transmitter emits a signal, that signal is much closer

to the device's antenna and therefore more powerful than any signal it receives. Expecting an antenna to both speak and listen at the same time is possible only with special echo-canceling technology [4].

1.5.6 Non-Orthogonal Multiple Access (NOMA)

All of the current cellular networks implement orthogonal multiple access (OMA) techniques such as time division multiple access (TDMA), frequency division multiple access (FDMA), or code division multiple access (CDMA) together. However, none of these techniques can meet the high demands of future radio access systems. In recent years, non-orthogonal multiple access (NOMA) schemes have received significant attention for the fifth generation (5G) cellular networks. The primary reason for adopting NOMA in 5G owes to its ability to serve multiple users using the same time and frequency resources. There exist two main NOMA techniques: power-domain and code-domain.¹ Power-domain NOMA attains

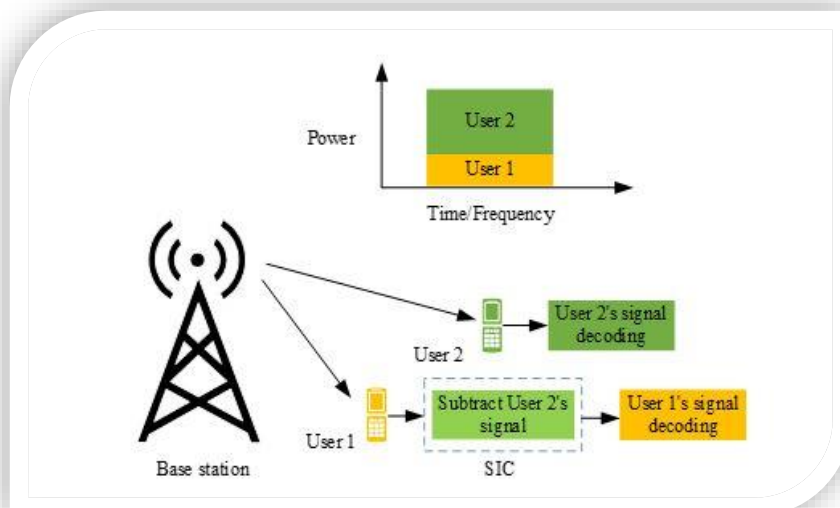


Figure 1.4 One BS two user with NOMA [5]

multiplexing in the power domain, whereas code-domain NOMA achieves multiplexing in the code domain [5]. NOMA exploits superposition coding at the transmitter and successive interference cancellation (SIC) at the receiver, thus multiplexing users in the power domain. As shown in Figure (1.4), the base station (BS) sends the superposed signals to two users, where User 1 has a higher channel gain than User 2. In NOMA, the user with higher channel

gain and the user with lower channel gain is usually referred to as the strong user and the weak user, respectively. The strong user first subtracts the signal of the weak user through SIC, and then decodes its own signal; the weak user considers the signal of the strong user as noise and detects its own signal directly. With worse channel gain and more interference, the weak user is assigned more power in NOMA to ensure fairness [5]. NOMA dominates orthogonal multiple access (OMA) in several aspects, such as:

- 1) It achieves superior spectral efficiency by serving multiple users at the same time and with the same frequency resource and mitigating the interference through SIC.
- 2) It increases the number of simultaneously served users, and thus, it can support massive connectivity;
- 3) Due to the simultaneous transmission nature, a user does not need to go through a scheduled time slot to transmit its information, and hence, it experiences lower latency;
- 4) NOMA can maintain user-fairness and diverse quality of service by flexible power control between the strong and weak users; particularly, as more power is allocated to a weak user, NOMA offers higher cell-edge throughput and thus enhances the cell-edge user experience. [6]

But there are various limitations and implementation issues that need to be addressed to exploit the full advantages of NOMA, such as:

- 1) Each user needs to decode the information of all other users with worse channel gains (which are in the same cluster) before decoding its own information leading to additional receiver complexity and energy consumption compared with OMA.
- 2) When an error occurs in SIC at a user, the subsequent decoding of all other users' information will likely be carried out erroneously. This implies keeping the number of users in each cluster reasonably low to reduce the effect of error propagation.

- 3) To obtain the claimed benefits of power-domain multiplexing, a considerable channel gain difference between the strong and weak users is required. This intuitively restricts the effective number of user pairs, which in turn reduces the sum-rate gain of NOMA.
- 4) Each user needs to send back its channel gain information to the BS, and NOMA is inherently sensitive to the uncertainty in the measurement of this gain [6].

1.5.7 Network Slicing

Logical network slices create tenant or service-specific networks. The network slicing creates end-to-end isolated logical networks starting from the mobile edge, continuing through the RAN mobile transport through the 5G core. Tenants are service providers delivering specific services over the network. These tenants will have specific network requirements such as reliability, latency, or bandwidth. Service-specific networks will have key performance indicator (KPI) requirements to meet a specific business need. In figure (1.5) the NFV and SDN technologies have been used to create four isolated and independent logical networks. The 5G network resources are “sliced.” Each network slice has different network performance specifications for a different use case or business requirements [7], and we will discuss more about the NFV and Network Slicing in chapter 2.

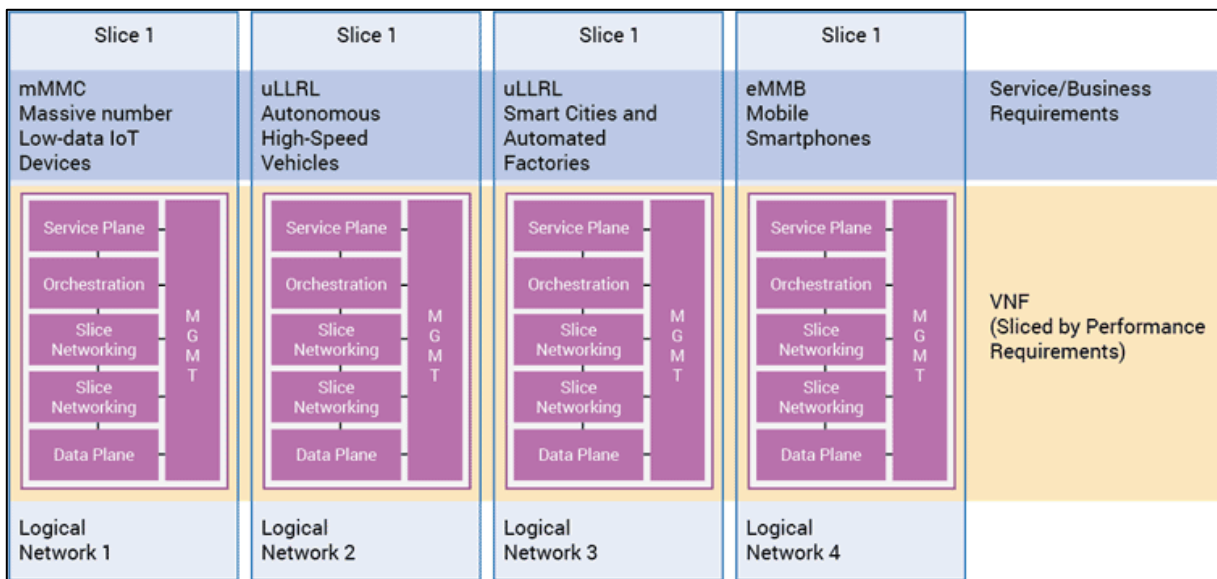
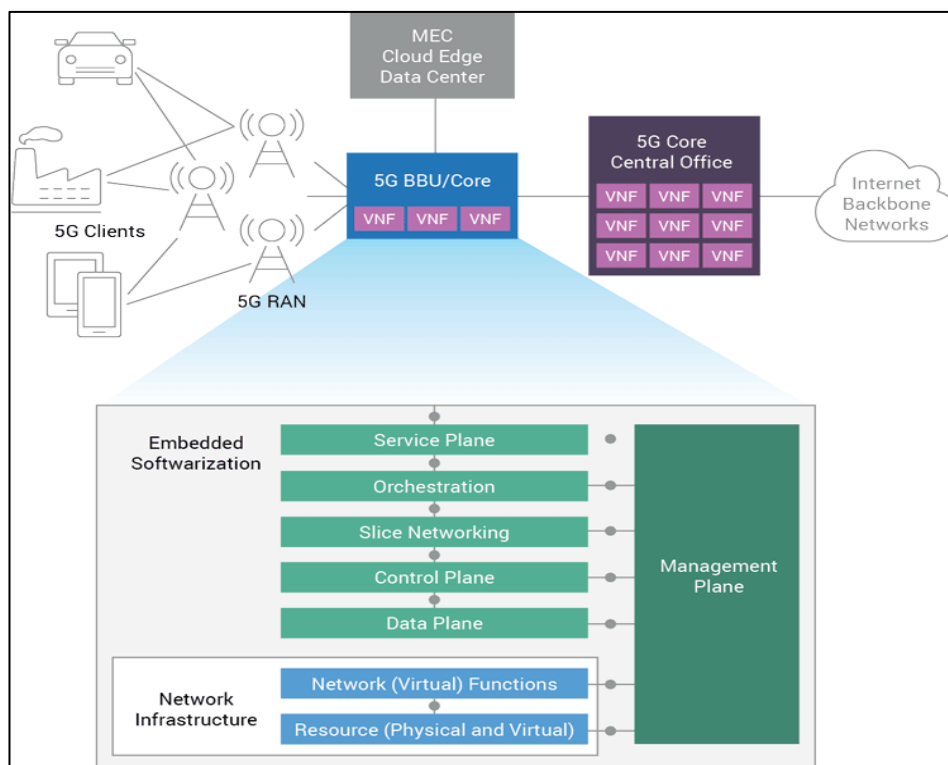


Figure 1.5 Network slicing example with-four logical network slices [7]

1.5.8 Infrastructure Based on NFV and SDN

One of the core architectural requirements specified for 5G infrastructures is that all core network systems are based on software virtualization. Network infrastructures have traditionally included physical purpose-built appliances, which are less flexible to manage, deploy and scale. Network Functions Virtualization (NFV) is a network architecture based on Virtual Network Functions (VNF). Network functions include network routing, packet processing, security, and many others. Software-Defined Networks (SDNs) are a virtualization technology that abstracts physical networks to virtual network structures. Virtual networks appear and behave like physical networks but have similar advantages of other virtualization technologies. In figure (1.6) shows a simplified view of 5G core functions using NFV/SDN. The traffic management including services, orchestration, control, and data packet management is implemented as a set of VNF chained services. The physical network resources are presented with a virtual network overlay using SDN [7].



1.5.9 Multi-

Figure Error! Use the Home tab to apply 0 to the text that you want to appear here..6 5G Packet Core processing functional overview [7]

Tenancy

Multi-tenancy uses network slicing and subscriber awareness to create isolated logical networks for independent service providers. Tenant networks can be defined with different performance characteristics and service levels. As in figure (1.7) the Tenant A has an isolated logical network provided by network slicing. Tenants A and B share a common physical network infrastructure [7].

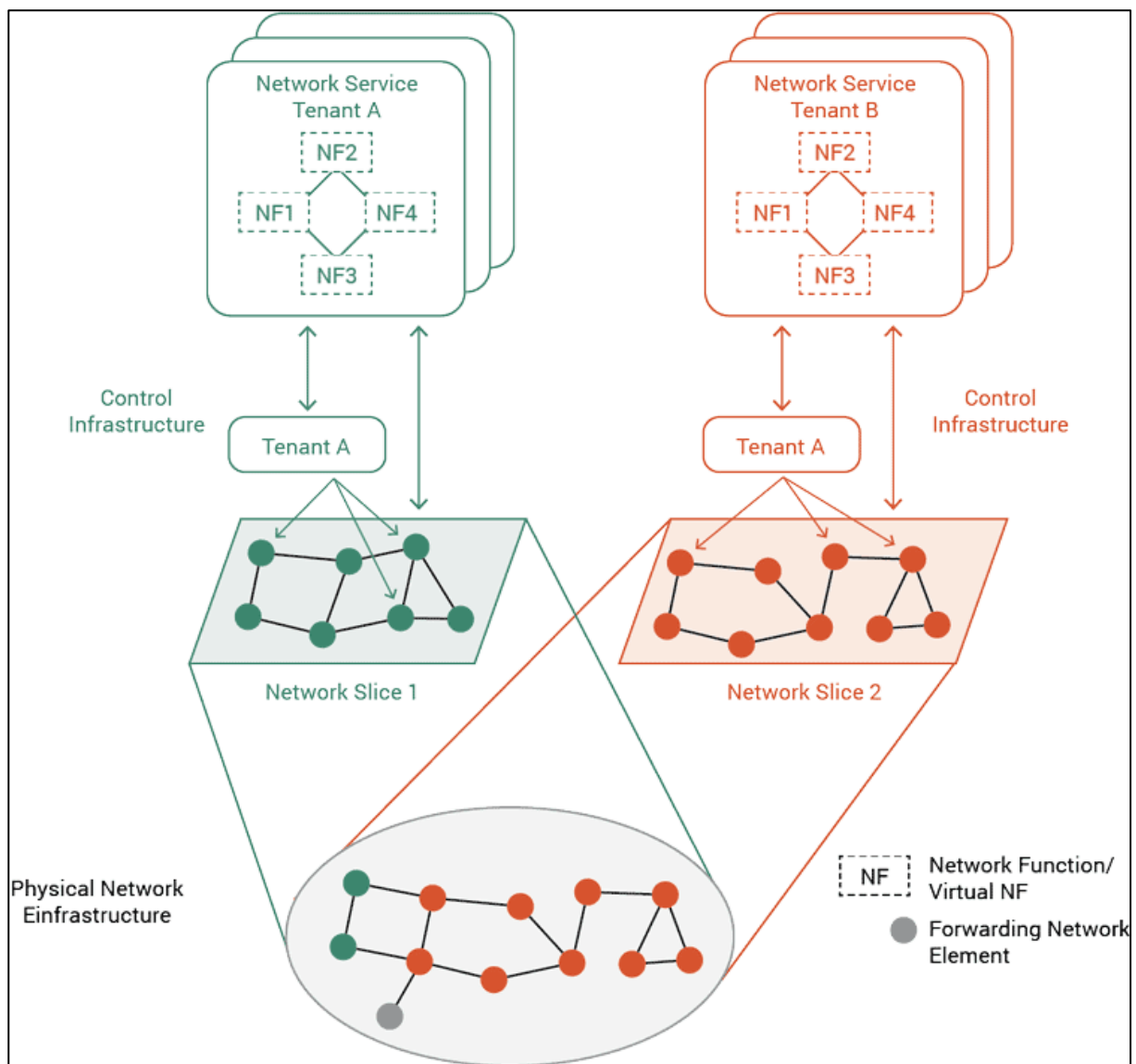


Figure 1.7 Multi-Tenancy [7]

Chapter 2

5G

Virtual Network

2.1 The concept of Network Slicing

5G networks are intended to provide mainly three types of services: massive machine type communications (mMTC), ultra-reliable low latency communications (URLLC), and enhanced mobile broadband (eMBB). The mMTC is characterized by massive number of devices communicating with each other and requires low cost along with long battery backup time. Apart from that, URLLC need simultaneous low latency and ultra-reliability. On the other hand, eMBB requires higher data rates along with large coverage area. To enable users with eMBB, mMTC, and URLLC, it's imperative to redesign the network architecture. Network slicing is a promising candidate to allow 5G networks to offer a wide variety of services and applications such as e-health, augmented reality, smart transportation system, smart banking, smart farming, and mobile gaming. The notion of the network slicing is to leverage the network infrastructure resources for creation of multiple subnetworks for different types of services and applications. Then, each subnetwork will perform slicing of the physical network resources to yield an independent network for its applications. To enable different 5G services, slices of different types for mMTC, URLLC, and eMBB services that utilize the network resources can be created. We can assign each slice type with complete end-to-end network resources; however, it does not seem practical because of the high expense. On the other hand, it would be feasible to allow sharing of the network resources among multiple types of slices using technologies like network function virtualization and software defined networking. The notion of network slicing is the virtual network architecture that will enable powerful and flexible capability of creating multiple logical networks on top of the common physical infrastructure as illustrated in Figure (2.1). Network softwarization is an emerging concept that will enable the network slicing using software-based solutions. Network softwarization can be achieved through technologies such as Network Function Virtualization (NFV) and Software Defined Networking (SDN). Specifically, network slicing in 5G will use SDN, NFV, cloud computing, and edge computing

to enable flexible operation for different types of services over the same physical infrastructure. Network slicing allows the creation of logical networks for different types of services as illustrated in Figure (2.1). Each logical network will have independent control with the ability of creation on demand [8].

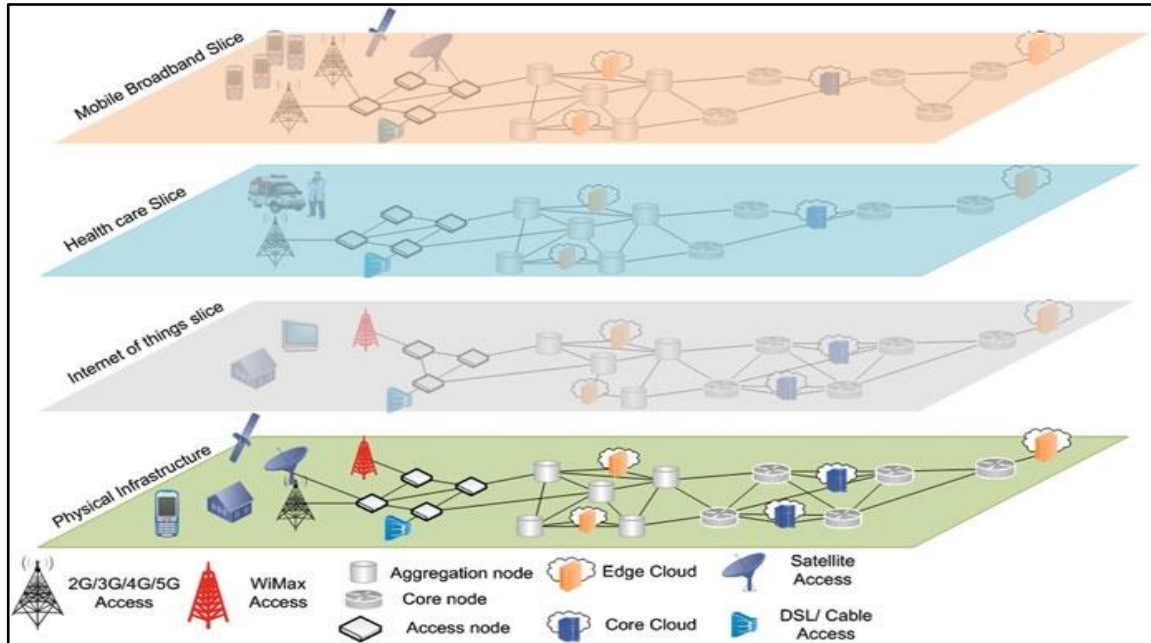


Figure 2.1 5G network slicing [8]

A generic framework for 5G network slicing is presented as shown in Figure (2.2). This framework has three layers such as service layer, network function layer, and infrastructure layer. Apart from these three layers, it contains Management and Orchestration (MANO) entity. The MANO performs the translation of service models and use cases into slices. The network infrastructure layer deals with the physical network infrastructure that covers both core network and radio access network. Other than this, the infrastructure layer also performs infrastructure control and allocation of resources to the slices. Apart from infrastructure layer, the network function layer performs the encapsulation of the operations required in network functions configuration and life cycle management. These network functions are then chained together to deliver an end-to-end service [8].

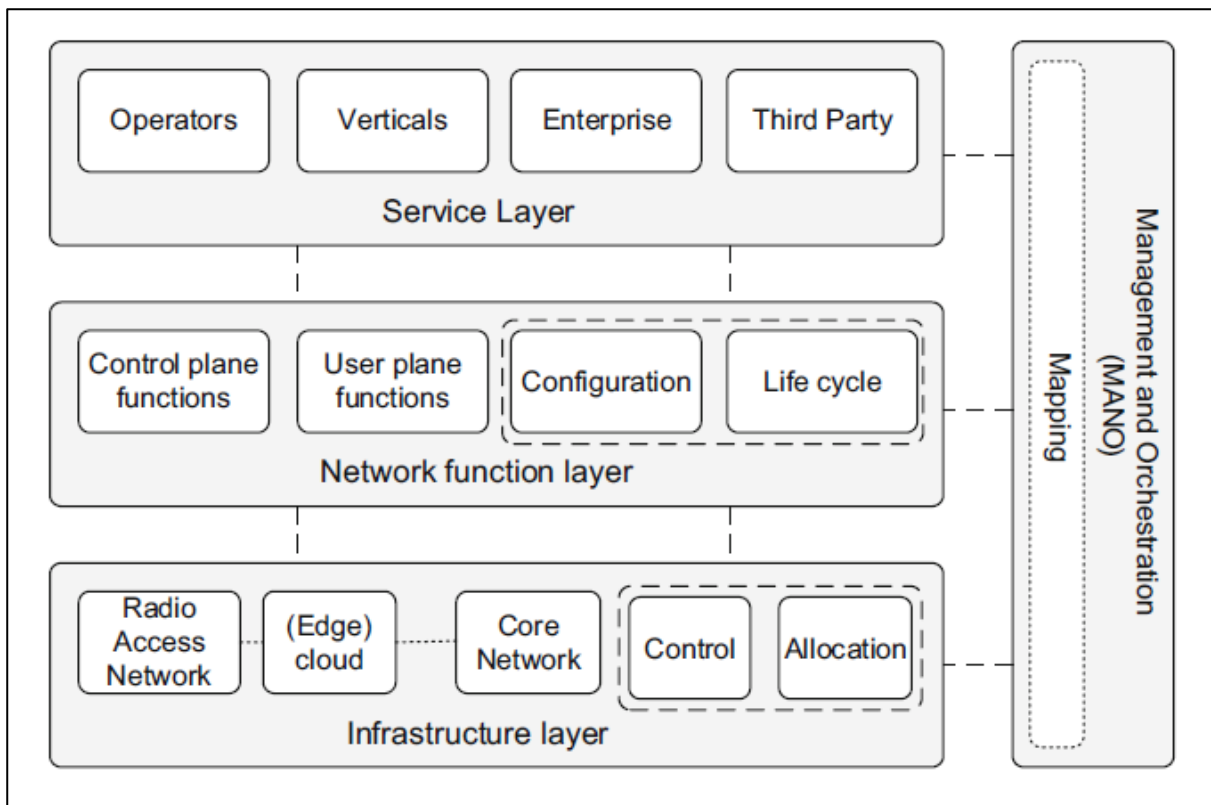


Figure 2.2 5G network slicing layered framework [8]

Following are the definitions of some terms that will be used later in this chapter.

Software Defined Networking: SDN allows the separation of control functionalities from the data transmission network. It actually divides the network into two planes such as data plane for data transmission and centralized control plane for performing control layer functionalities. The main advantage of software defined networking is the easier management due to its centralized nature; however, the centralized controller might suffer from the inefficiency when the scale of the network increases [8].

Network Function Virtualization: NFV is envisioned to yield the architecture that allows virtualization of the functionalities of the network nodes, which are then chained together for performing different communication services. A virtual network function might comprise of one or more virtual machines running on different devices such as servers, edge computing nodes, cloud computing infrastructure, and network switches.

Tenant: Tenant denotes the users that can access the shared resources with specific privileges and access rights.

Infrastructure Providers: It refers to operators that provide the physical infrastructure and are responsible for maintenance and operation of the network.

2.2 Network Slicing Principles

Network slicing principles enable the creation of multiple logical networks on top of the common physical network infrastructure to provide services to different market applications.

Network slicing should be done according to the following three main principles.

2.2.1 Slice Isolation

Slice isolation is one of the important properties that should be guaranteed in performing network slicing. It guarantees the performance with security of different tenants by enabling the feature that one tenant slices are independent of the other slices. Apart from that, isolation makes it possible to restrict the tenant user in accessing/modifying the other tenant slices. Isolation is a basic characteristic of the network function virtualization with additional capability of imposing limits on the network resources usage. This feature enables guaranteeing the performance of different tenants by fair sharing of the resources [8].

2.2.2 Elasticity

Elasticity allows the dynamic alteration of resources allocated to different tenants slices in order to effectively utilize the resources. Elasticity can be realized by relocating the virtual network functions, scaling up/down of the allocated resources, and reprogramming the control and data elements functionalities. The main challenge that will exist in the implementation of the elasticity is the policy for interslice negotiation so that the performance of different slices remains unaffected [8].

2.2.3 End-to-End Customization

Customization makes sure that allocated shared resources to different tenants are utilized effectively. It is expected that customization will be offered in future by many vertical industries. To enable customization, network function virtualization, software defined networking, and network orchestration can be exploited for providing resource allocation in an agile way [8].

2.3 Network Slicing Enablers

Network slicing in 5G will be enabled by software defined networking, network function virtualization, cloud computing, and edge computing. 5G architecture is envisioned to provide slices of different services such as eMBB services, URLLC services, and mMTC services. This section discusses the key enablers of the network slicing.

2.3.1 Software Defined Networking

Traditional networking devices were designed to perform a set of functionalities and vendors need to update the design to enable new features. One solution is to build a custom hardware; however, this approach has disadvantage of high cost associated with building of custom hardware. Other than cost, significant efforts will be needed to come up with design that meet the most updated specifications. Once the system is designed, there is no way to make changes in it. All of the above factors motivated researchers to develop novel architecture that will enable configurable feature. SDN is a solution that is envisioned to enable users with separation of control plane from the underlying network devices, more granular security, lower operating costs, and reduced capital expenditure. The scale of the network evolution imposes challenges on its management and configuration in a traditional network. SDN centralized architecture enables easier management of the large-scale networks by decoupling the control plane from

data plane as shown in Figure (2.3). The logical control plane has the ability of easier configuration along with effective management functions. In software defined networking, the job of forwarding packets is performed by the centralized network controller through programmable interfaces [8]. OpenFlow is the most commonly used interface for controlling the packet flow [8]. The rise in scale of the network imposes stress on the centralized control in SDN. This has limitation effect on the OpenFlow networks scalability. To minimize load on the centralized controller, it is preferable to design an architecture that enables the processing of frequent events near the switches. A hierarchical architecture is proposed to reduce the load on the centralized controller. The controllers are divided into two categories such as local and root controllers. The local controller is connected to one or more switches, which is further

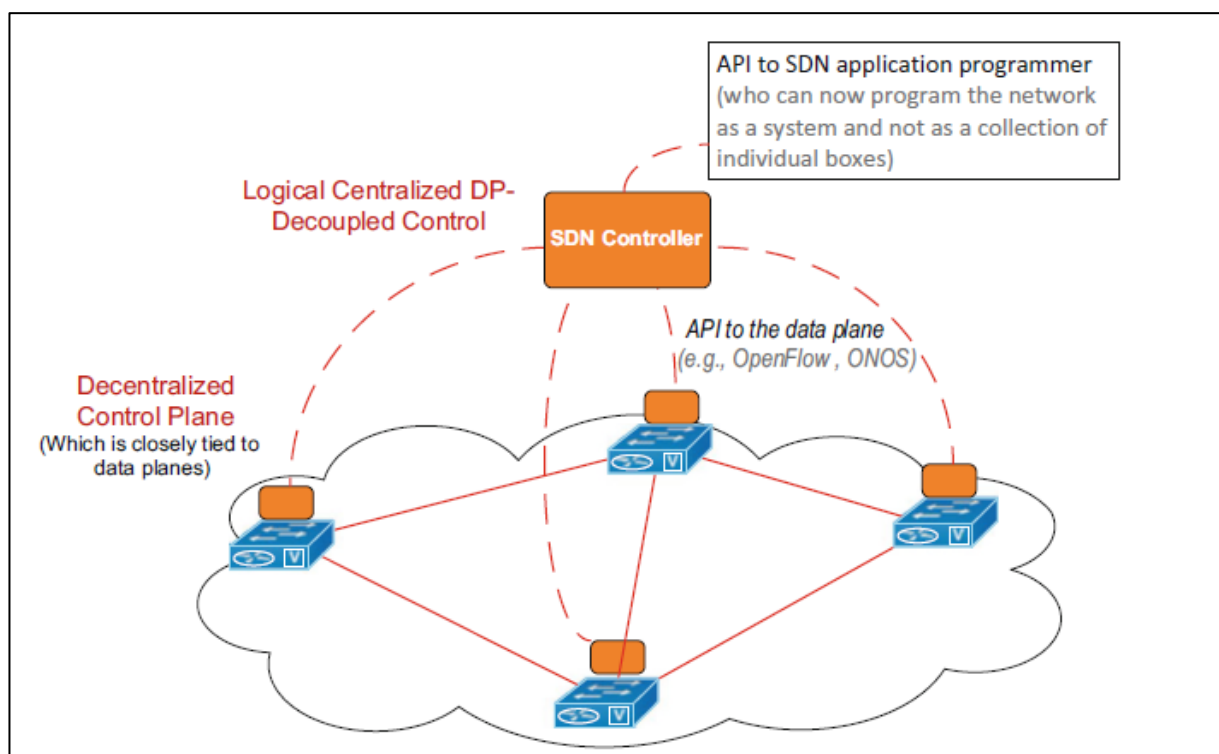


Figure 2.3 Software defined networking [8]

controlled by the root controller. The root controller other than controlling the local controller performs functions that need the network wide view of the network.

2.3.2 Network Function Virtualization

Network function virtualization (NFV) is a concept that leverages virtualization feature for transforming the network nodes functions into virtual functions that can be further chained together to enable different communication services. One or more virtual machines that run on

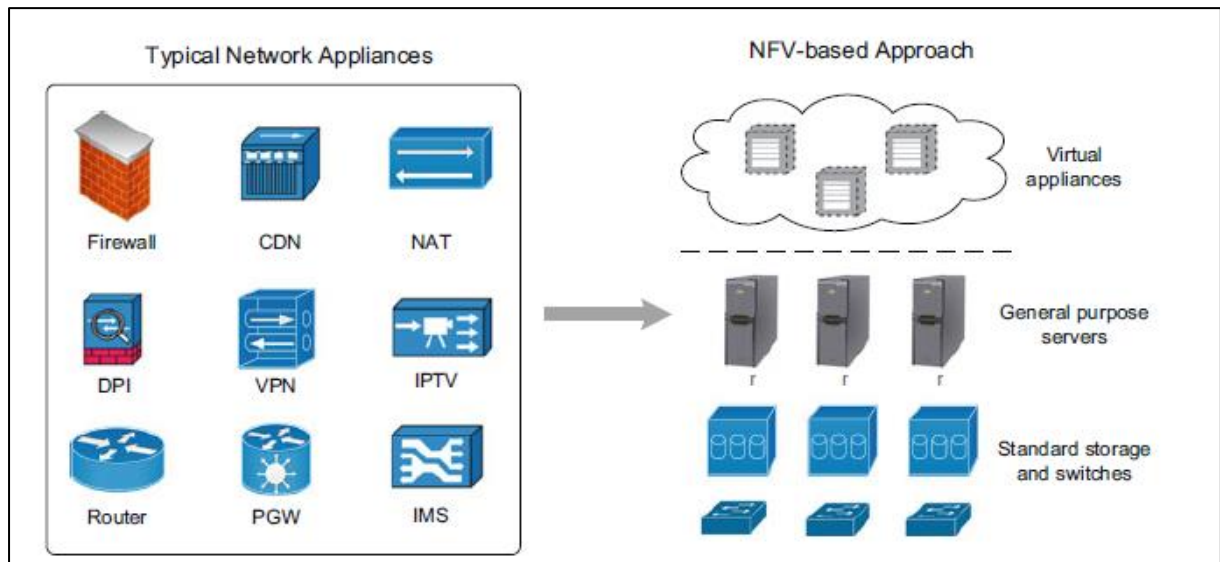


Figure 2.4 NFV-based approach of instantiating network functions [8]

different network nodes such as network switches and servers can be used to run a virtual network function. NFV uses commodity hardware to run software virtualization techniques for implementation of network functions as shown in Figure (2.4).

The virtual appliances have the advantage of instantiating without installation of new hardware.

NFV will enable operators with the following advantages:

1. Reduction of the capital expenditure by enabling operators with network functions without buying the hardware and payment as per usage.
2. It also allows the operators to reduce the operational expenditure required for space, cooling, and power.
3. It adds more flexibility to the system by enabling easier and quick scaling of the services.

In the context of network slicing, the framework of network function virtualization enables management of virtual network functions, service chaining, and latency oriented virtual network function embedding. The architecture of network function virtualization given in consists of three main components: network function virtualization orchestrator, virtual network function manager, and virtualized infrastructure manager. The virtual network function manager performs the life cycle management of instances of virtual network functions. The virtual infrastructure manager is responsible for management and control of resources of network function virtualization infrastructure of network operator. The network function virtualization orchestrator performs orchestration of the network resources, validation and authorization of the virtual network function manager requests for resource of network function virtualization infrastructure, and life cycle management of the network service [8].

2.3.3 Cloud Computing

Cloud computing often called as cloud enables users with on-demand online pool of shared resources that include computing and storage resources, respectively. The cloud follows three types of delivery models such as infrastructure as a service (IAAS), platform as service (PAAS), and software as service (SAAS). SAAS allows users with on-demand use of softwares without installing it on a local personal computer. The software runs on cloud that is available for users and the users only pay for the software as per the use in a cheap way. Further, the users can access the software regardless of the location and enjoy more freedom of working in a group of peoples located at different places. The cloud software can be accessed either via client application or a browser. Examples of SAAS are dropbox, Gmail, Microsoft office 365, Facebook, etc. Apart from SAAS, PAAS allows the users with platform for development of applications and is made up of a database, browser, operating system, and programming language execution environment. PAAS enable the users to develop their applications regardless of taking care of the underlying infrastructure which is managed by the vendor.

Examples of PAAS include Apprenda, windows Azure, Amazon web services, Engine Yard, AWS Elastic Beanstalk, and force.com. In IAAS, the virtualized computing resources are offered as a service via internet. Examples of IAAS include Amazon EC2, Gogrid, DigitalOcean, Cisco Metapod, and Google Compute Engine [8].

the requirements for cloud computing architecture in 5G can be divided into three categories such as provider requirements, enterprise requirements, and user requirements

Reliability: One of the most important requirements is the reliability that must be fulfilled in the cloud system. The cloud computing system might suffer from hardware and software flaws due to which it will not work properly. The system should be designed in way that must be resilient to both type of flaws because of the fact that some applications (such as vehicular communication and e-health) in 5G require ultra-high reliability.

Sustainability: Sustainability deals with the use of renewable energy source and energy efficient hardware design. In 5G, we will witness a massive number of devices that will access the network, therefore, it's imperative to design the sustainable system that will fulfill the users requirements without adding pollution to the environment.

Scalability: It deals with the management of increase in complexity due to an increase in computation tasks. Scalability in clouds can be either horizontal or vertical. Horizontal scalability is provided through load balancing. On the other hand, vertical scaling is to scale up capacity through increase in server resources. Most common way to do vertical scaling is to use expensive hardware and then utilize the hardware as a virtual machine hypervisor.

Fault Tolerance: Fault tolerance is the ability of a cloud computing system to continue its operation if some of its component stops working. The cloud service provider should design the fault tolerant system that detect failure on time and then, apply mechanisms to recover from the outage within short time [8].

2.3.4 Edge Computing

Although cloud computing provides the advantages of scalability, elasticity, resource pooling, and sufficient storage capacity, the high latency associated with cloud makes it unattractive for real time and resource-intensive applications. To enable users with resource-intensive and latency sensitive applications such as augmented reality and e-health, edge computing comes into the picture which leverages pushing of the storage and computing resources to the edge of the network. In literature, three different terms such as cloudlets, fog computing, and mobile edge computing appeared till now and are based on pushing of the computing and storage resources to the network end. All of these terms represent edge computing with different context awareness, scope, and edge nodes nature features, respectively. The term cloudlet was coined by Satyanarayana et al. to enable the placement of small-scale data center at the network edge for execution of resource-intensive applications. Fog computing was coined by CISCO to refer to architecture that uses the edge devices (such as routers and gateways) to allow users with computing and storage capabilities. The term mobile edge computing was coined by European Telecommunications Standards Institute (ETSI) to propose the placement of servers at the base stations [8].

Chapter 3

Resource Allocation

3.1 General introduction

In economics, resource allocation is the assignment of available resources to various uses. In the context of an entire economy, resources can be allocated by various means, such as markets, or planning where as in project management, resource allocation or resource management is the scheduling of activities and the resources required by those activities while taking into consideration both the resource availability and the project time and in networks resource allocation is an important feature meant to ensure high efficiency for networks in many resources as in energy, bandwidth, data rate and other, as well as its maintenance as a cost-benefit network. Proper resource allocation improves the performances of both the associated system and the network, and also helps in avoiding the different kinds of transient bottlenecks involved in the network.

3.2 Resource allocation in cloud radio access 5G network

3.2.1 Introduction

Cloud radio access network (CRAN) is considered as an important paradigm to enhance the performance of the upcoming fifth generation (5G) and beyond wireless networks in terms of capacity, latency, and connectivity to a large number of devices. Out of several potential enablers, efficient resource allocation can mitigate various challenges related to user assignment, power allocation, and spectrum management in a CRAN and we will talk in this chapter about a various aspects of resource allocation. More importantly, we identity and discuss the key elements for efficient resource allocation and management in CRAN, namely: user assignment, remote radio heads (RRH) selection, throughput maximization, spectrum management, network utility, and power allocation. Typically, in a CRAN, the resource allocation tasks are performed at the centralized entity BBU and the RRHs are only used for transmission and reception of the signal. Since the BBU pool has complete information about

the connected RRHs in the network, the resource allocation task can easily be executed there. However, managing the resource allocation task with various conflicting objectives (resource, throughput and energy) and constraints (power, throughput, resource, quality, CPU, cloud, and memory) is a challenging task and it must be done while maintaining the quality of service (QoS) parameters of various network entities [9].

3.2.2 The resource allocation element in a CRAN

3.2.2.1. User assignment

Like in any wireless network, efficient user assignment is of vital importance in a CRAN. User scheduling refers to the selection of a group of users in a particular time slot with the objective of enhancing the network throughput. Concerning the limited resources and interference constraints, intelligent scheduling of users is the key for enhance the throughput of the network, and for subsequently minimizing the impact of interference. On the other hand, evaluating the group of users for a particular time slot in a CRAN is a computationally intense task and is one of the active research areas nowadays [9].

3.2.2.2 RRH selection

In a CRAN, the functionality of RRH is simply the transmission and reception of the signal. However, the RRH selection is a critical task and it has a direct impact on the spectral and energy efficiency of the network. In addition, RRHs can cooperate with each other and perform the centralized beamforming task which can also have direct impact on the throughput enhancement of the wireless channels [9].

3.2.2.3 Throughput

To meet the ever-increasing throughput demand in 5G and beyond wireless networks, the CRAN framework can greatly complement the enhancement in the throughput. More specifically,

many techniques can be adopted to enhance the throughput and meet the ever-growing demand including network coding, beamforming, and power management [9].

3.2.2.4 Spectrum

Spectrum management is key for getting the real benefits of CRAN technology. Integration of licensed and unlicensed bands at the RRH is one of the active research areas nowadays. To achieve this, an optimized framework for the integration should be designed in such a way that the performance of legacy Wi-Fi users should not be compromised. On the other hand, mmWave band is also proposed as an important ingredient for 5G and beyond wireless networks. It is also suggested that mmWave is more appropriate for the backhaul link part. For the successful deployment of the mmWave band, spectrum management is a key research area for getting the real benefits from it [9].

3.2.2.5 Network utility

Network utility is a general term and it encompasses various QoS parameters including throughput, delay, outage and blocking probabilities. The main concern in designing the network utility is to guarantee minimum QoS requirements of network entities [9].

3.2.2.6 Power allocation

Efficient power allocation is of vital importance in any wireless network. In a CRAN, it is more challenging because the RRHs are close apart in the 5G and beyond networks and this creates significant interference problems. In addition, efficient power allocation is also important because it has a direct impact on the energy and spectral efficiency [9].

3.2.3 Emerging use-cases with the CRAN architecture

In this section, we discuss several emerging use-cases and Radio Access Technologies (RATs) in 5G and beyond wireless networks, which can be supported by the CRAN architecture.

3.2.3.1 Heterogeneous CRAN (H-CRAN)

Beyond 5G wireless networks are envisioned to be highly heterogeneous in terms of access technologies and device capabilities, and highly dense due to the deployment of small/femto/pico cells to enhance the cellular capacity and coverage, leading to huge prevalence of HetNet. The ever-increasing cellular densification will enable the paradigm shift of the existing networks with the features of offloading, coverage expansion, capacity enhancement, and user quality of experience (QoE) improvement. On the other hand, CRAN can be a promising platform to manage the ultra-dense HetNets including macro and small/pico/femto BSs, thus resulting in the concept of H-CRAN. In contrast to the conventional CRAN (with a centralized processing unit and a large number of geographically separated small cells), H-CRAN also involves macro cells and allow the option of decoupling user-plane and control plane. With this functionality splitting approach, small cells may deal with the user traffic while macro cells may handle the signaling traffic, i.e., control signaling to maintain connectivity within a large area. Such a splitting helps in reducing signaling traffic over the fronthaul and also to enhance the resource-usage efficiency, energy efficiency, and overall QoE [9].

3.2.3.2 Millimeter wave (mmWave) CRAN

Cellular network densification and mmWave communications are considered promising enablers to meet the data rate requirements of future beyond 5G networks by utilizing highly dense BSs/access points (APs) and huge bandwidth available in the higher frequency bands, respectively. Also, CRAN can enable the network densification in a cost-efficient manner for enhancing the spectral efficiency and energy efficiency of the next-generation wireless networks. By integrating CRAN architecture with the mmWave communications, both the objectives of huge bandwidth and network densification can be achieved, thus leading to the concept of mmWave CRAN. Furthermore, in contrast to the communications in low-frequency

bands, mmWave communications is more susceptible to blockages including shadowing and building walls and is also affected by atmospheric conditions [9].

3.2.3.3 NOMA-based CRAN

NOMA has been considered as a promising multiple access technology for future wireless networks and it has been already included in the 3GPP LTE-A standard for improving the spectral efficiency and energy efficiency. The combination of NOMA with CRAN can be a potential solution for the IoT-enabled resource-constrained wireless networks and NOMA-enabled RAN framework can enhance the performance of cell-edge users. The key technologies identified for NOMA-enabled H-CRANs include massive MIMO, cognitive radio, mmWave communications, wireless charging, cooperative transmission, and device-to-device (D2D) communications [9]. we will discuss more about D2D resource allocation in other section.

3.2.3.4 Full-duplex enabled CRAN

With the recent advances in self-interference (SI) mitigation techniques including analog cancellation, digital cancellation and antenna cancellation, full-duplex (FD) technology has emerged as a promising technique to enhance the spectral efficiency in 5G and beyond wireless networks. However, the main challenge for implementing FD-enabled cellular system is the mutual interference between uplink and downlink transmissions. Although various multi-user detection/decoding techniques have been suggested for interference mitigation in FD-enabled CRAN, their implementation is complex from a practical perspective [9].

3.2.3.5 Cell-free massive MIMO

The massive antenna array utilized in massive MIMO-enabled BSs can be deployed either in collocated or distributed set-ups. As compared to the collocated massive MIMO which has the benefits of low back-haul bandwidth requirements, the distributed massive MIMO systems can provide much higher probability coverage at the cost of increased backhaul bandwidth by

exploiting the diversity against shadow fading. In the distributed multi-antenna setting, the recently emerging concept of cell-free massive MIMO comprises of a huge number of distributed BSs/APs which concurrently serve a much smaller number of users distributed over a wide area. In such a distributed system set-up, all the BSs/APs connected to a centralized unit cooperate phase-coherently over a backhaul to serve all users with the same frequency-time resources and there are no boundaries or cells, thus referred as “cell-free massive MIMO”. Similar to the benefits highlighted earlier for the case of mmWave-based CRAN, mmWave technology can be utilized in combination with cell-free massive MIMO systems to exploit the high bandwidth at the mmWave frequencies. Due to high-power consumption levels and production costs at the mmWave frequencies, fully digital implementation of massive MIMO systems is challenging and to address this, hybrid transceiver architectures with both the analog and digital processing and hybrid beamforming/precoding techniques seem promising

3.2.3.6 Virtualized CRAN

Another emerging CRAN architecture is a virtualized CRAN which can be enabled with the virtualization and software defined networking paradigms. The network function virtualization (NFV) enables sharing of various resources such as licensed spectrum, backhaul, core and access network, power and network infrastructure among MNOs for better resource utilization efficiency and energy efficiency. Furthermore, the combination of CRAN and virtualization paradigm helps to mitigate the issues of existing LTE-based networks including dynamic traffic management and information exchange among cells in dense cellular networks [9]. We will discuss more about resource allocation for virtualized networks in other section.

3.3 Resource allocation in D2D

Resource management includes interference management and power control. The resource allocation approaches can be centralized, distributed or semi-distributed.

In the centralized approach, the Base station is responsible for the allocation and management of the resources. The complexity of the centralized approach is more as the Base Station which is responsible for monitoring the performance of the system, SINR and channel quality calculation, connection establishment, and call setup should also allocate the resources and control the interference in the system. This complexity also grows as the number of users in the network increases as the Base Station without any assistance has to collect the necessary information from the devices and measure the quality of the channel and the performance of the network. Therefore, the centralized approach is only suitable for small cell networks and is more complex for larger networks. The distributed approach, however, does not involve the Base Station in the management of resources. The devices are responsible for the allocation and management of resources. This type is also suitable for larger networks with a greater number of devices. The devices gather information about the neighboring devices more often by the exchange of messages. The devices that intend to communicate in the D2D mode monitors the cellular resources and opportunistically access the licensed resources of cellular communication. The devices monitor cellular communication to gather information about the channel quality, SINR and also about the availability of cellular resources. Resource allocation is also a solution to avoid undesirable interference in a communication network. Minimization of interference is done by an efficient allocation of resources like spectrum and power which is the most challenging issue in D2D communication [10].

3.4 Resource allocation in virtual networks

Virtualization of a network is considered as one of the essential formative view-points in the pushing toward fifth generation (5G) cellular networks. Many inherent benefits such as enhanced data rates, spectrum/energy efficiency, capacity, and lower latency are expected to be achieved for higher quality of experience. However, the unprecedented growth in data traffic and the tsunami of mobile devices in the existing networks demands to consider spectrum

efficiency and massive connectivity in 5G networks. Also, the greatest limitation of the existing employed access scheme, i.e., orthogonal multiple access (OMA) scheme (i.e., the OFDMA scheme) is that the number of users that can simultaneously occupy the spectrum resources is restricted by the number of available spectrum resources, i.e., subchannels, resource blocks (RBs). Recently, a new promising technology has been under consideration for solving the challenges of OMA-based schemes. Non-orthogonal multiple access (NOMA) has been viewed as a key enabler for catering the inconveniences of OFDMA scheme in 5G networks. NOMA takes advantage of the resource gain differences through which it allows multiple users to be scheduled on a single spectrum resource. Thus, the NOMA technique can accommodate higher number of users in the network when compared to traditional OMA schemes. However, users that are scheduled over the same spectrum resource create inter-user interference over the spectrum resource. This problem can be solved by using the successive interference cancellation (SIC) technique at the receiver. However, reaping the benefits of NOMA requires meeting significant challenges in terms of resource and power assignment [8].

3.4.1 Resource allocation in network slicing

The concept of network slicing (NS) has been proposed for flexible resource provisioning where a physical resource is partitioned into logically independent networks on demand. The NS resource allocation implies the definition of a feasible path in the infrastructure network with adequate resource availability. However, due to complex structural characteristics of the backhaul transport network, a number of issues arise when fast deploying the end-to-end (E2E) slices onto network infrastructures. In order to improve extreme quality of service (QoS) and user experience, latency-optimal virtual resource allocation problem will be defined, subject to the backhaul capacity and bandwidth constraints. The concept of network slicing (NS) is considered as an efficient solution to address the diverse requirements of applications in 5G networks. By slicing a single physical substrate network, multiple end-to-end (E2E) logical

networks are tailored for operators to provide corresponding services in parallel, where the NS is mutually isolated, managed independently and created on demand. Each logical slice corresponds to an abstraction of a subset of physical substrate network resources, aiming to logically separate the set of virtual network functions (VNFs) within the physical infrastructure. For example, one network slice is dedicated to augmented reality applications with ultra-reliable and low-latency communications, whereas another slice is designated for extremely high throughput video-on-demand services. With the powerful software-defined networking (SDN) and network function virtualization (NFV) technologies available, NS can be centrally managed and fast deployed with better resource utilization and cost efficiency according to the requirement of each use case. With NFV, it would be possible to flexibly tailor the slice for a specific scenario, while SDN controller is acknowledged as an enabler to remotely configure the physical network. Although the topic of enabling technologies, SDN and NFV, are well investigated for many slice realizations models, the extension to the characteristics of NS (i.e., the topology, required virtual functions and interconnections) is not straightforward. In order to provide efficient and practical deployment policies for E2E slices in the backhaul transport network, it is of major significance to construct the logical NS model, substrate network model, and resource mapping relationships. In the logical network, each NS can be interpreted as an E2E traffic flow with multiple communication service instances to support specific and extreme performances in terms of latency, throughput, capacity and availability. Each communication service instance is composed of a sequence of virtual network units (VNUs) and virtual links, which can be illustrated as a service chain (SC). VNUs which carry dedicated VNFs (e.g., BBUs (Building Baseband Units), GWs (Gateways) or MMEs (Mobility Management Entities)) can be deployed onto network data centers (DCs) and run on the general-purpose hardware. A virtual link between VNUs can be realized as a multi-hop physical path. Hence, the NS resource allocation can be defined as a feasible path that slice traffic should follow in infrastructure networks with adequate resource availability. In other words, the realization of NS, in essence,

is to deploy VNUs and virtual links into physical communications infrastructures. Although the concept of NS is still nascent, NFV and SDN paradigms for achieving NS have been maturely researched on concrete solutions and readily available platforms. By logically partitioning physical network resources in an efficient manner, NFV resolves many problems facing NS. For example, NFV can tailor the slice for a specific scenario and assign specific network functions to each slice flexibly, constructing dynamic and service-aware networks with lower operating and capital expenses. Because of the novelty of NS, many challenges in aspects, e.g., wireless slicing, slice resource allocation with guaranteed extreme QoS and isolation in multi-domain networks remain to be addressed. In the context of wireless networks, resource allocation has captured much attention due to significant challenges in energy efficiency (EE), bandwidth allocation, power control and so on. There are significant efforts in the network EE optimization under the QoS constraints and a transmit power budget [11].

Chapter 4

Optimization

4.1 Optimization Problem Types Overview

In an optimization problem, the types of mathematical relationships between the objective and constraints and the decision variables determine how hard it is to solve, the solution methods or algorithms that can be used for optimization, and the confidence you can have that the solution is truly optimal.

A key issue is whether the problem functions are convex or non-convex. If the objective and all constraints are convex, we can be confident of determining whether there is a feasible solution, finding the globally optimal solution, and solving the problem up to very large size. If any functions are non-convex, the problem is much harder and you cannot be certain of any of these things. Below is a list of optimization problem types, arranged in order of increasing difficulty for the solution methods:

- Linear and Quadratic Programming Problems.
- Quadratic Constraints and Conic Optimization Problems.
- Integer and Constraint Programming Problems.
- Smooth Nonlinear Optimization Problems.
- Nonsmooth Optimization Problems.

4.1.1 Convex and Non-Convex Optimization's Problem

In fact, the great watershed in optimization isn't between linearity and nonlinearity, but convexity and nonconvexity. Convex optimization problems are far more general than linear programming problems, but they share the desirable properties of LP problems: They can be solved quickly and reliably up to very large size hundreds of thousands of variables and constraints. The issue has been that, unless your objective and constraints were linear, it was

difficult to determine whether or not they were convex. A convex optimization problem is a problem where all of the constraints are convex functions, and the objective is a convex function if minimizing, or a concave function if maximizing. Linear functions are convex, so linear programming problems are convex problems. Conic optimization problems the natural extension of linear programming problems are also convex problems. In a convex optimization problem, the feasible region the intersection of convex constraint functions is a convex region, as pictured below.

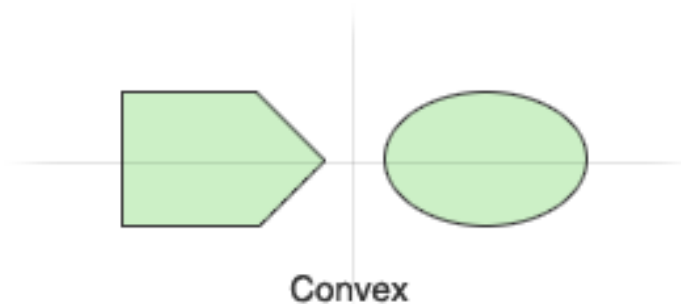


Figure 4.2 convex problem representation [13]

With a convex objective and a convex feasible region, there can be only one optimal solution, which is globally optimal. Several methods notably Interior Point methods will either find the globally optimal solution, or prove that there is no feasible solution to the problem. Convex problems can be solved efficiently up to very large size.

A non-convex optimization problem is any problem where the objective or any of the constraints are non-convex, as pictured below.

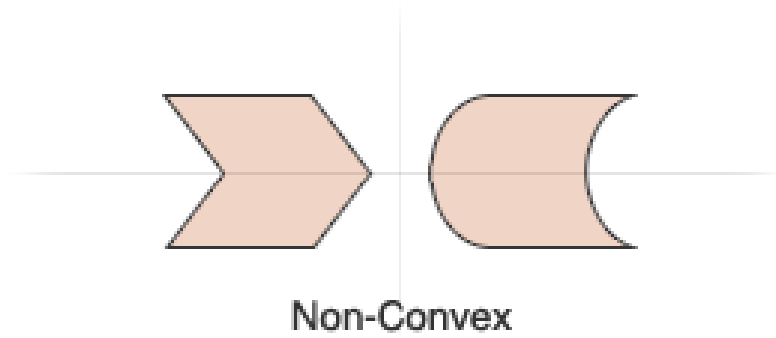


Figure 4.2 non-convex problem representation [13]

Such a problem may have multiple feasible regions and multiple locally optimal points within each region. It can take time exponential in the number of variables and constraints to determine that a non-convex problem is infeasible, that the objective function is unbounded, or that an optimal solution is the "global optimum" across all feasible regions.

4.1.2 Convex and Non-Convex Functions

Geometrically, a function is convex if a line segment drawn from any point $(x, f(x))$ to another point $(y, f(y))$ called the chord from x to y lies on or above the graph of f , as in the picture below:

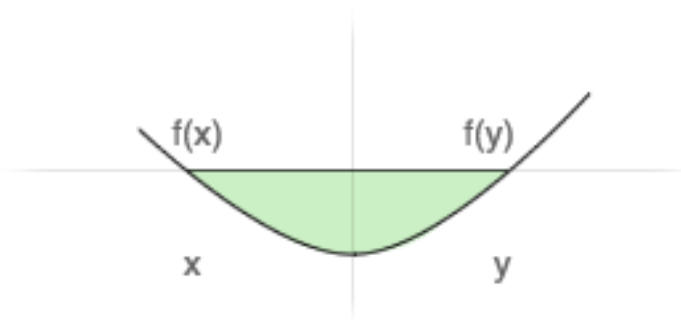


Figure 4.3 A convex function [13]

Algebraically, f is convex if, for any x and y , and any t between 0 and 1, $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$. A function is concave if f is convex i.e., if the chord from x to y lies on or

below the graph of f . It is easy to see that every linear function -- whose graph is a straight line is both convex and concave.

A non-convex function “curves up and down” it is neither convex nor concave. A familiar example is the sine function:

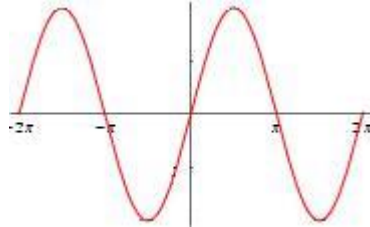


Figure 4.4 Non-convex function [13]

but note that this function is convex from $-\pi$ to 0 , and concave from 0 to $+\pi$. If the bounds on the variables restrict the domain of the objective and constraints to a region where the functions are convex, then the overall problem is convex.

Therefore, we’ve chosen our optimization problem’s type to be convex, where the optimal solution can be simply obtained.

4.2 Optimizing Problem

4.2.1 Problem Overview

The problem of resource allocation is modeled as an optimizing problem. The ability to solve the optimization problem requires that one entity must have all information related to the optimization problem. In our system model, this entity is the cloud server. Therefore, the cloud server observes the overall system and makes the decisions of task assignment periodically and based on the system state. When a new task arrives at one MU node, the MU node sends a request to the assigned eNodeB, which in its turn forward the request to the cloud server.

Considering the system state and the available resources on the eNodeBs, the cloud server handles the request and assigns the new task to one eNodeB based on the task's requirements.

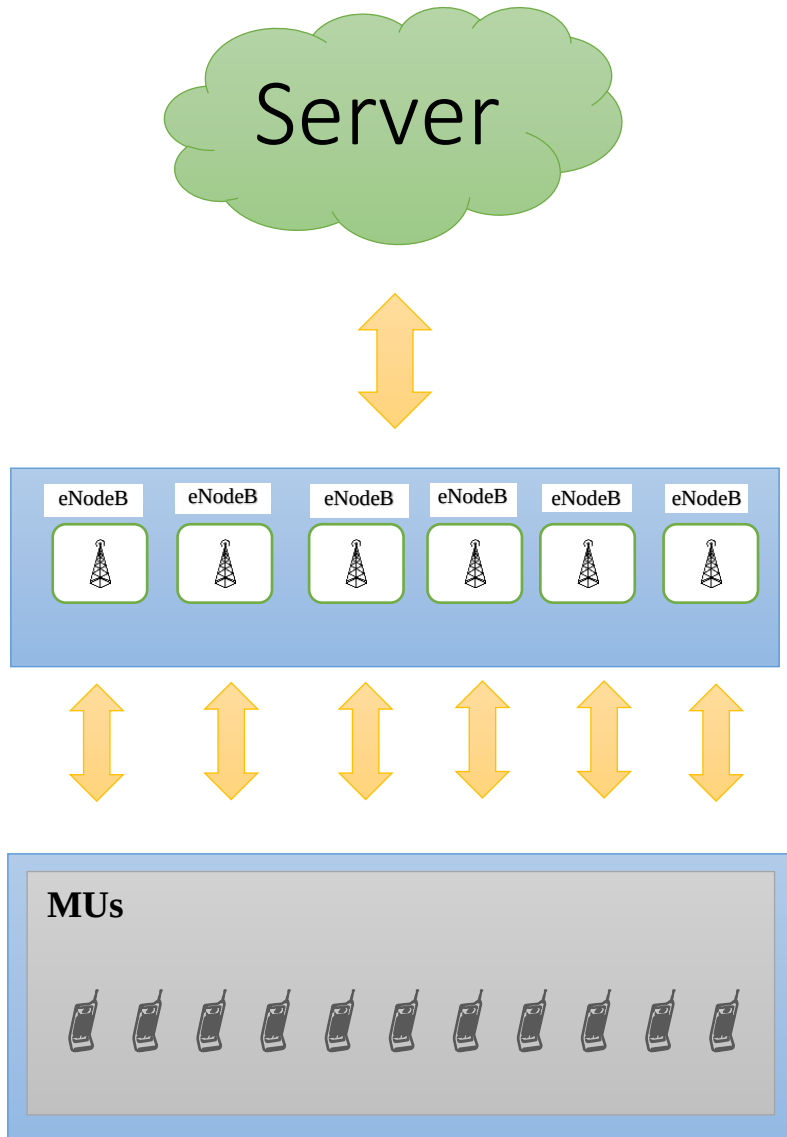


Figure 4.5 Network architecture

There are N MU nodes in the system that form the set of MU nodes $\mathcal{T}$. These MU nodes are connected to a set $\mathcal{F}$ of nearby eNodeBs, where the number of eNodeBs is $F = |\mathcal{F}|$. Each MU is referred to as $MU_i \in \mathcal{T}$ and is assigned to one eNodeB $f_j \in \mathcal{F}$ determined by the server. There is no direct connection between the server and MU nodes, where the communication is done

via eNodeBs. MU nodes periodically generate new tasks which are offloaded to an eNodeB that is specified by the server. $task_{ij}$ refers to a task generated by the MU node IoT_i and is being served by a number η_i of VMs on the eNodeB f_j . Each eNodeB operates a number N_{VM} of virtual machines (VMs), where each virtual machine (VM) can handle at most one task, while a task may require multiple VMs to be handled. We refer to the number of VMs needed to handle task i as η_i . Without loss of generality, we assume that all eNodeBs have the same number K of VMs, while the number of occupied VMs differentiates. We refer to the available number of VMs on eNodeB j as A^j .

4.2.2 Problem Formulation

The problem is to minimize the difference in resources occupation among eNodeBs. The aim is to make the occupancy ratio of the eNodeBs as equal as possible, bypassing scenarios in which some eNodeBs are fully occupied while others are nearly idle. In other words, in solving this problem, we aim to achieve load balance between the eNodeBs. This optimization problem has some constraints, such as that the tasks are serviced within the time limit for each task.

First, we define the following binary variables as a step towards the formulation. y_k^j indicates whether a VM number k on eNodeB j is occupied by a task or not:

$$y_k^j = \begin{cases} 1: & \text{if VM number } k \text{ on fog node } j \text{ is occupied} \\ 0: & \text{Otherwise} \end{cases} \quad (1)$$

The variable $x_k^{i,j}$ indicates whether the task generated by MU node i and assigned to eNodeB j is mapped to VM number k or not:

$$x_k^{i,j} = \begin{cases} 1: & \text{if } task_{ij} \text{ is assigned to VM number } k \\ 0: & \text{Otherwise} \end{cases} \quad (2)$$

The relation between $x_k^{i,j}$ and y_k^j is determined as:

$$\sum_{i=1}^N x_k^{i,j} = y_k^j; \forall k = 1, 2, \dots, K; \forall j = 1, 2, \dots, F; \quad (3)$$

where, one VM can be assigned to at most one task.

Therefore, the number of occupied VMs on eNodeB j is computed as:

$$A^j = \sum_{k=1}^K y_k^j; \forall j = 1, 2, \dots, F; \quad (4)$$

The total number of occupied VMs over all eNodeBs is:

$$A = \sum_{j=1}^F A^j. \quad (5)$$

Consequently, the average occupancy of VMs over all the eNodeBs is calculated as follows:

$$\bar{A} = \frac{A}{F.K}. \quad (6)$$

Since $\bar{A}$ represents the average occupancy of VMs over all the eNodeBs, the aim would be to bring the occupancy of all eNodeBs close to this amount. In other words, we need to minimize the occupancy variance which is the square of difference between the average occupancy and the occupancy of each eNodeB. Therefore, the occupancy variance of the eNodeB j can be obtained as:

$$var(j) = (\bar{A} - A^j)^2 \quad (7)$$

Lastly, the objective function can be summarized as:

$$\min_{A^j} \sum_{j=1}^F (\bar{A} - A^j)^2 \quad (8)$$

The constraints related to the optimization problem are described below:

Each VM can serve one task at most.

$$\sum_{i=1}^N \sum_{j=1}^F x_k^{i,j} \leq 1; \forall k = 1, 2, \dots, K \quad (9)$$

The number of VMs needed to handle task i as η_i

$$\sum_{j=1}^F \sum_{k=1}^K x_k^{i,j} = \eta_i; \forall i = 1, 2, \dots, N \quad (10)$$

The relation between $x_k^{i,j}$ and y_k^j is determined as:

$$\sum_{i=1}^N x_k^{i,j} = y_k^j; \forall k = 1, 2, \dots, K; \forall j = 1, 2, \dots, F; \quad (11)$$

Assigning a VM to one task requires that this VM becomes occupied:

$$x_k^{i,j} \leq y_k^j; \forall i = 1, 2, \dots, N; ; \forall k = 1, 2, \dots, K; ; \forall j = 1, 2, \dots, F;$$

The constraint related to the total number of VMs on each eNodeB:

$$\sum_{i=1}^N \sum_{k=1}^K x_k^{i,j} \leq \sum_{k=1}^K y_k^j; \forall j = 1, 2, \dots, F; \quad (12)$$

Therefore, the optimization problem is defined as:

$$\min_{A^j} \sum_{j=1}^F (\bar{A} - A^j)^2 \quad (13a)$$

Subject to:

$$\sum_{i=1}^N \sum_{j=1}^F x_k^{i,j} \leq 1; \forall k = 1, 2, \dots, K \quad (13b)$$

$$\sum_{j=1}^F \sum_{k=1}^K x_k^{i,j} = \eta_i; \forall i = 1, 2, \dots, N \quad (13c)$$

$$\sum_{i=1}^N x_k^{i,j} = y_k^j; \forall k = 1, 2, \dots, K; \forall j = 1, 2, \dots, F; \quad (13d)$$

$$x_k^{i,j} \leq y_k^j; \forall i = 1, 2, \dots, N; ; \forall k = 1, 2, \dots, K; ; \forall j = 1, 2, \dots, F; \quad (13e)$$

$$\sum_{i=1}^N \sum_{k=1}^K x_k^{i,j} \leq \sum_{k=1}^K y_k^j; \forall j = 1, 2, \dots, F; \quad (13f)$$

$$x_k^{i,j} \in \{0,1\}; \forall i = 1, 2, \dots, N; ; \forall k = 1, 2, \dots, K; ; \forall j = 1, 2, \dots, F; \quad (13g)$$

$$y_k^j \in \{0,1\}; \forall k = 1, 2, \dots, K; ; \forall j = 1, 2, \dots, F; \quad (13h)$$

It is clear that the problem is convex, where the optimal solution can be simply obtained. To exploit the optimization problem traditionally, we need to solve the problem whenever a new

task is generated at one MU node. To overcome this shortage, we propose the following algorithm to assign only the new task to a suitable eNodeB, while the previous assignments stay the same. To do this, we need to count the current occupied VMs over all eNodeBs.

In the run-time operation, when a new task is generated at one MU node, the task's requirements are forwarded to the server. The server has a complete database which contains the available resources on each eNodeBs.

We assume that in each specific time period a number of tasks are generated by MU nodes. The server is aware of these tasks and it distributes them to the appropriate eNodeBs, taking into account that there are tasks that are currently being executed and that are served by the eNodeBs. The server distributes these tasks to the eNodeBs by solving the optimization problem to achieve a load balance between the eNodeBs.

Chapter 5

Network Simulation

Using OMNeT++

5.1 Introduction to OMNeT++

OMNeT++ is an object-oriented modular discrete event network simulation framework. It has a generic architecture, so it can be (and has been) used in various problem domains:

- modeling of wired and wireless communication networks
- protocol modeling
- modeling of queueing networks
- modeling of multiprocessors and other distributed hardware systems
- validating of hardware architectures
- evaluating performance aspects of complex software systems
- in general, modeling and simulation of any system where the discrete event approach is suitable, and can be conveniently mapped into entities communicating by exchanging messages.

OMNeT++ itself is not a simulator of anything concrete, but rather provides infrastructure and tools for writing simulations. One of the fundamental ingredients of this infrastructure is a component architecture for simulation models. Models are assembled from reusable components termed modules. Well-written modules are truly reusable, and can be combined in various ways like LEGO blocks. Modules can be connected with each other via gates (other systems would call them ports), and combined to form compound modules. The depth of module nesting is not limited. Modules communicate through message passing, where messages may carry arbitrary data structures. Modules can pass messages along predefined paths via gates and connections, or directly to their destination; the latter is useful for wireless simulations, for example. Modules may have parameters that can be used to customize module behavior and/or to parameterize the model's topology. Modules at the lowest level of the module hierarchy are called simple modules, and they encapsulate model behavior. Simple modules are

programmed in C++, and make use of the simulation library. OMNeT++ simulations can be run under various user interfaces. Graphical, animating user interfaces are highly useful for demonstration and debugging purposes, and command-line user interfaces are best for batch execution. The simulator as well as user interfaces and tools are highly portable. They are tested on the most common operating systems (**Linux**, Mac OS/X, Windows), and they can be compiled out of the box or after trivial modifications on most Unix-like operating systems. OMNeT++ also supports parallel distributed simulation. OMNeT++ can use several mechanisms for communication between partitions of a parallel distributed simulation, for example MPI or named pipes. The parallel simulation algorithm can easily be extended, or new ones can be plugged in. Models do not need any special instrumentation to be run in parallel it is just a matter of configuration. OMNeT++ can even be used for classroom presentation of parallel simulation algorithms, because simulations can be run in parallel even under the GUI that provides detailed feedback on what is going on. OMNEST is the commercially supported version of OMNeT++. OMNeT++ is free only for academic and non-profit use; for commercial purposes, one needs to obtain OMNEST licenses from Simulcraft Inc [12].

5.2 Download and install Omnet++

```
C:\Program Files\WSL\wsl.exe
OMNeT++ Installation Environment - opp_env@0.35.0.260420

To see all the supported simulation models, run:
  opp_env list

To create a new workspace and install the latest version of OMNeT++, run:
  mkdir workspace && cd workspace && opp_env init && opp_env install omnetpp-latest && opp_env shell omnetpp-latest

To install and use the latest version of the INET Framework with a single command, run:
  opp_env shell --install inet-latest

To run a simulation model directly with the latest version of OMNeT++, run:
  opp_env run --install omnetpp-latest -c 'cd $OMNETPP_ROOT/samples/aloah;./aloah'

Run 'touch ~/.opp_env_intro_disable' to remove this intro.

INFO Using specified projects [inet-4.6.0] with effective projects [inet-4.6.0, omnetpp-6.3.0] in workspace /home/opp_env

INFO Downloading project omnetpp-6.3.0 in workspace /home/opp_env
https://github.com/omnetpp/omnetpp/releases/download/omnetpp-6.3.0/omnetpp-6.3.0-linux-x86_64.tgz
===== 100.0%

INFO Downloading project inet-4.6.0 in workspace /home/opp_env
https://github.com/inet-framework/inet/releases/download/v4.6.0/inet-4.6.0-src.tgz
===== 100.0%

INFO Patching project inet-4.6.0
Enabling feature(s): ZGateSchedulingConfigurator
  opp_featuretool: 'opp_featurestate' updated
  opp_featuretool: 'nedexclusions' updated
Enabling feature(s): VoipStream, VoipStreamComplex

omnetpp@0.35.0.260420:~$ ls
Simu56-1.2.2  inet-4.5.2-src.tgz  inet4.5  omnetpp-6.0.3  omnetpp-6.0.3-linux-x86_64.tgz
omnetpp@0.35.0.260420:~$ cd omnetpp-6.0.3/
omnetpp@0.35.0.260420:~/omnetpp-6.0.3$ .setenv
Environment for 'omnetpp-6.0.3' in directory '/home/omnetpp-6.0.3' is ready.

Type './configure' and 'make' to build the simulation libraries.
When done, type 'omnetpp' to start the IDE.
omnetpp@0.35.0.260420:~/omnetpp-6.0.3$ omnetpp
omnetpp: command not found
omnetpp@0.35.0.260420:~/omnetpp-6.0.3$ ./configure
configure: Environment variables (PATH and PYTHONPATH) are correctly set.
configure: Reading configure.user for your custom settings.
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for clang... no
checking for gcc... no
checking for g++... no
checking for clang... no
checking for cc... no
checking for alx... no
configure: error: in '/home/omnetpp-6.0.3':
configure: error: no acceptable C compiler found in $PATH
See 'config.log' for more details
omnetpp@0.35.0.260420:~/omnetpp-6.0.3$
```

```
C:\Program Files\WSL\wsl.exe
OMNeT++ Installation Environment - opp_env@0.35.0.260420

To see all the supported simulation models, run:
  opp_env list

To create a new workspace and install the latest version of OMNeT++, run:
  mkdir workspace && cd workspace && opp_env init && opp_env install omnetpp-latest && opp_env shell omnetpp-latest

To install and use the latest version of the INET Framework with a single command, run:
  opp_env shell --install inet-latest

To run a simulation model directly with the latest version of OMNeT++, run:
  opp_env run --install omnetpp-latest -c 'cd $OMNETPP_ROOT/samples/aloah;./aloah'

Run 'touch ~/.opp_env_intro_disable' to remove this intro.

omnetpp@0.35.0.260420:~/default_workspace$ ls
omnetpp@0.35.0.260420:~/default_workspace$ cd ..
omnetpp@0.35.0.260420:~$ ls
default_workspace  inet-4.6.0  omnetpp-6.3.0
omnetpp@0.35.0.260420:~$ cd omnetpp-6.3.0/
omnetpp@0.35.0.260420:~/omnetpp-6.3.0$ .setenv
Environment for 'omnetpp-6.3.0' in directory '/home/omnetpp-6.3.0' is ready.
omnetpp@0.35.0.260420:~/omnetpp-6.3.0$ omnetpp
<D> Error: This OMNeT++ installation cannot be used outside an opp_env shell.
omnetpp@0.35.0.260420:~/omnetpp-6.3.0$ opp_env shell omnetpp-6.3.0

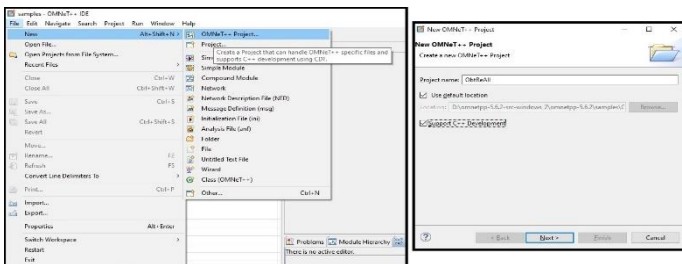
INFO Using specified projects [omnetpp-6.3.0] with effective projects [omnetpp-6.3.0] in workspace /home/opp_env

INFO Starting non-isolated shell for projects [omnetpp-6.3.0] in workspace /home/opp_env
Warning: overwriting previous environment for '/home/opp_env/omnetpp-6.3.0'.
Environment for 'omnetpp-6.3.0' in directory '/home/opp_env/omnetpp-6.3.0' is ready.
HINT To build, clean, test or check a project or all projects, use the following commands: 'build_all', 'clean_all', 'test_all', 'smoke_test_all', 'check_all'; 'build_omnetpp', 'clean_omnetpp', 'test_omnetpp', 'smoke_test_omnetpp', 'check_omnetpp'. (Use 'declare -f' commands to check what they do.)
omnetpp@0.35.0.260420:~/omnetpp-6.3.0$
omnetpp@0.35.0.260420:~/omnetpp-6.3.0$ omnetpp
Starting the OMNeT++ IDE...
omnetpp@0.35.0.260420:~/omnetpp-6.3.0$
```

5.3 Setting up the project in OMNet++

To build our practical project for optimizing resource allocation in 5G network we start to build the architecture of the network on a project in OMNeT++, first we create a project named () as shortcut for optimizing resource allocation as showing in figure (5.1).

After this an empty project will be created, this project will hold all files that belong to our simulation.



5.4 Adding omnetpp.ini file

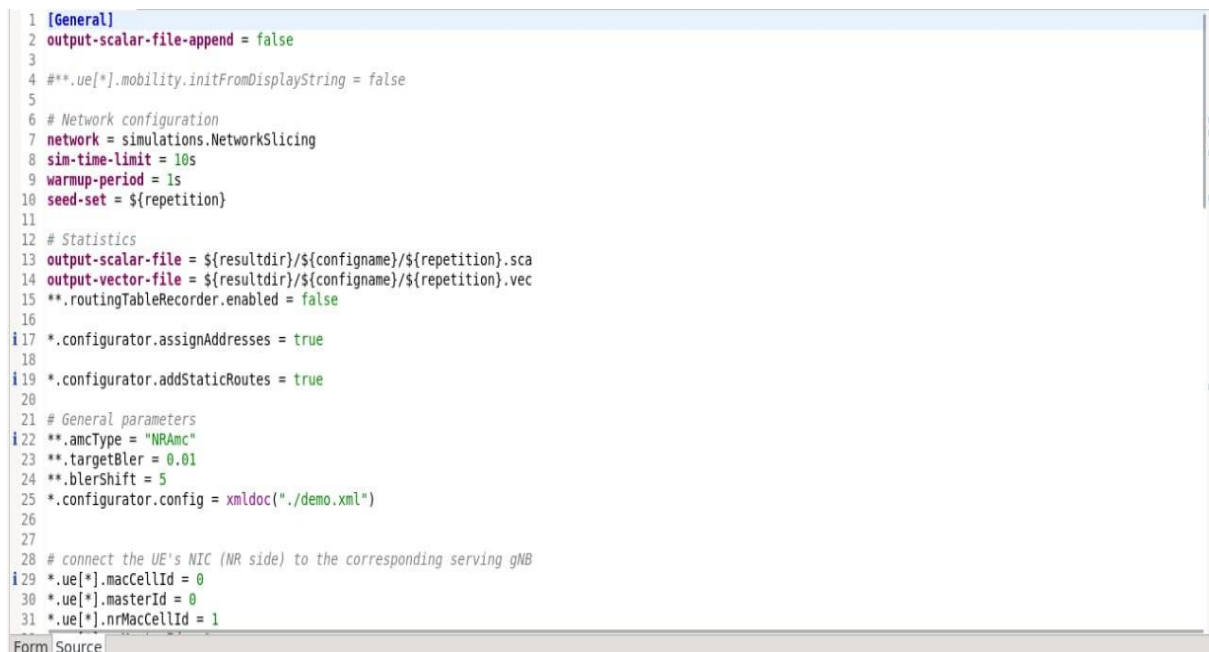


Figure 5.4.1

```

32 *.ue[*].nrMasterId = 1
33
34 # Application Setup
35 *.ue[*].numApps = 1
36 *.server.numApps = 1
37
38 *.ue[*].app[*].typename = "CbrReceiver"
39 *.ue[*].app[*].localPort = 1000
40
41 *.server.app[*].typename = "CbrSender"
42 *.server.app[*].PacketSize = 1000 # send 800 Kb/s
43 *.server.app[*].sampling_time = 0.01s
44 *.server.app[*].destAddress = "ue["+string(ancestorIndex(0))+"]" # obtain the address of the client by reading its index in the array of Apps
45 *.server.app[*].destPort = 1000
46 *.server.app[*].localPort = 3000+ancestorIndex(0)
47 *.server.app[*].startTime = uniform(0.01s,0.05s)
48
49 # ????? ???? ?????? ???? ????? ???? ?????????? (????? 1000x1000 ???)
50 *.ue[*].mobility.constraintAreaMinX = 0m
51 *.ue[*].mobility.constraintAreaMaxX = 2000m
52 *.ue[*].mobility.constraintAreaMinY = 0m
53 *.ue[*].mobility.constraintAreaMaxY = 2000m
54 *.ue[*].mobility.constraintAreaMinZ = 0m
55 *.ue[*].mobility.constraintAreaMaxZ = 0m # ?????? 0 ?????? ?????? ?? ?????? ???
56
57 [Config Single-UE]
58 *.numUe = 1
59
60 *.ue[*].mobility.typename = "StationaryMobility"
61
62 *.ue[*].mobility.initialX = uniform(0m, 1000m)

```

Form Source

Figure 5.4.2

```

62 *.ue[*].mobility.initialX = uniform(0m, 1000m)
63 *.ue[*].mobility.initialY = uniform(0m, 1000m)
64 *.ue[*].mobility.initialZ = 0m
65
66
67
68
69 [Config Mobility]
70 *.numUe = 2
71 # ????? ???? ?????? ??????
72 *.ue[*].mobility.typename = "RandomWaypointMobility"
73
74
75 *.ue[*].mobility.initialX = uniform(0m, 1000m)
76 *.ue[*].mobility.initialY = uniform(0m, 1000m)
77
78
79 # ?????? ?????? ??????
80 *.ue[*].mobility.speed = uniform(1mps, 10mps)
81
82
83
84
85
86
87
88
89
90
91
92

```

Form Source

Figure 5.4.3

5.5 Adding the Network Description File (NED)

To complete the description of our network we should define the MU and the eNodeB as a NED file and in the NED file for each element we can define a lot of parameters and properties in the figure (5.5.1) and the figure (5.5.2) we Illustrates the code which define our element.

In the section parameters in the code, we define some of the properties of our element such as the send and receive rate, and in the gates section, we declare that our element has a wireless port and can connect with other elements or devices wirelessly.

For sure the NED file is not enough to define the functionality of our element and we need to implement the functionality of the MU and eNodeB module in C++.

```
1 package simulations;
2
3
4 import inet.networklayer.configurator.ipv4.Ipv4NetworkConfigurator;
5 import inet.networklayer.ipv4.RoutingTableRecorder;
6 import inet.node.ethernet.Eth10G;
7 import inet.node.inet.Router;
8 import inet.node.inet.StandardHost;
9 import ned.IdealChannel;
10 import simu5g.common.binder.Binder;
11 import simu5g.common.carrierAggregation.CarrierAggregation;
12 import simu5g.nodes.Upf;
13 import simu5g.nodes.NR.gNodeB;
14 import simu5g.nodes.NR.NRUE;
15 import simu5g.nodes.backgroundCell.BackgroundCell;
16 import simu5g.world.radio.LteChannelControl;
17
18
19 network NetworkSlicing
20 {
21     parameters:
22         int numUe = default(1);
23         int numBgCells = default(0);
24         @display("bgb=2000,2000");
25     submodules:
26         configurator: Ipv4NetworkConfigurator {
27             @display("p=278.2686,101.80559");
28         }
29         binder: Binder {
30             @display("p=274.8751,332.5649");
31         }
32 }
```

Figure 5.5.1

```

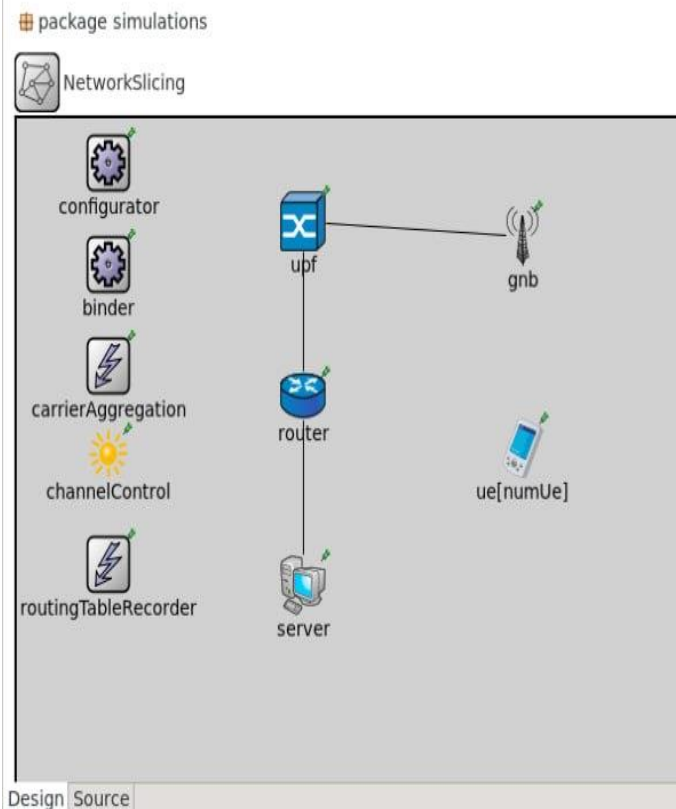
31 }
32 carrierAggregation: CarrierAggregation {
33   @display("p=274.8751,563.3242");
34 }
35 upf: Upf {
36   @display("p=858.5604,234.15285");
37 }
38 gnb: gNodeB {
39   @display("p=1516.9032,268.08804");
40 }
41 server: StandardHost {
42   @display("p=858.5604,1062.1716");
43 }
44 ue[numUe]: NRUE {
45   @display("p=1516.9032,749.96783");
46 }
47 channelControl: LteChannelControl {
48   @display("p=274.8751,760.1484");
49 }
50 routingTableRecorder: RoutingTableRecorder {
51   @display("p=274.8751,1014.66235");
52 }
53 router: Router {
54   @display("p=858.5604,631.19464");
55 }
56
57 connections:
58 upf.pppg++ <--> Eth10G <--> gnb.ppp;
59 upf.pppg++ <--> Eth10G <--> router.pppg++;
60 server.ethg++ <--> Eth10G <--> router.ethg++;
61 }

```

Design Source

Figure 5.5.2

5.7 Initial Network Simulation Scenario



<https://www.orange-business.com/sites/default/files/5g-usecase-insight.jpg>

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- [12] <https://doc.omnetpp.org/omnetpp/manual/>
- [13] <https://www.solver.com/problem-typ>

